How-To Library

Informatica

Automating the Configuration of a Generic JDBC Connection for Dynamic Data Masking

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Abstract

You can use a generic JDBC connection with Dynamic Data Masking. This article describes how you can automate the configuration of a generic JDBC client to work with Dynamic Data Masking. It includes the steps required to configure SQLWorkbench and includes three deployment tools that you can use. You can use the information as a reference for other database clients and configuration deployment tools.

Supported Versions

- Dynamic Data Masking 9.9.2
- Dynamic Data Masking 9.9.1
- Dynamic Data Masking 9.9.0
- Dynamic Data Masking 9.8.4

Table of Contents

Overview. 2
Rules and Guidelines. 3
Example. 3
Create an Installer Package. 3
Create and Package Required Files. 4
Using Microsoft Endpoint Configuration Manager to Deploy the Installer. 6
Using Other Deployment Management Tools. 23
Using Microsoft Endpoint Manager to Deploy the Installer. 23
Using Active Directory Group Policy Object to Deploy the Installer. 31

Overview

You can use Dynamic Data Masking to mask data in a source database that uses a generic JDBC connection.

You use a generic database client to access the database. Before you use a generic JDBC client, you must configure the client. The client machine must have the required JDBC drivers in the correct location.

Based on the number of users that access the database, multiple machines in an organization might require configuration. Managing the updates on individual machines can be difficult and might lead to issues.

You can automate the process of configuring client machines to ensure that all client machines have the required files and configurations. To automate the configuration process, create and deploy an installer that copies the Informatica JDBC drivers and adds the required configurations.

To deploy the installer, you can use a deployment management tool such as Microsoft Endpoint Configuration Manager or any deployment tool the organization uses to deploy required system updates.

Rules and Guidelines

Consider the following rules and guidelines when you automate the configuration of a generic database client:

- Install the database client in the same location on all client machines.
  The path must be the same because you specify the target directory when you deploy the installer. The installer uses the same path on all machines.
- Do not use the fresh install option on a machine that contains the configuration files. The fresh install option assumes that no files are present and does not replace or update files if they are present.
- Deploy the installer for one client installation per machine. You cannot use the installer to configure more than one client on a machine.

Example

An organization that uses Dynamic Data Masking creates generic JDBC connections and uses the SQLWorkbench client to access the database.

Because multiple users access the database, multiple machines require driver installation, maintenance, and regular updates to the latest drivers to access the database. Additionally, new users and machines are added in periodically and require the same updates.

The organization creates an installer to automate the process of configuring client machines.

Configuration of an SQLWorkbench client includes the following changes:

- Copy the Informatica JDBC JAR file to the client machine.
- Create an `SQLWorkbench64_ddm.ini` file with the required entries.
- Create an `SQLWorkbench64_ddm.exe` file to use to access the database through Dynamic Data Masking.

To deploy the installer to all client machines, the organization uses Microsoft Endpoint Configuration Manager which is used to deploy regular security updates.

Create an Installer Package

You can use an installer authoring tool, such as Microsoft Installer (MSI), to create the installer for SQLWorkbench.

Create an installer with options for fresh install and upgrade.

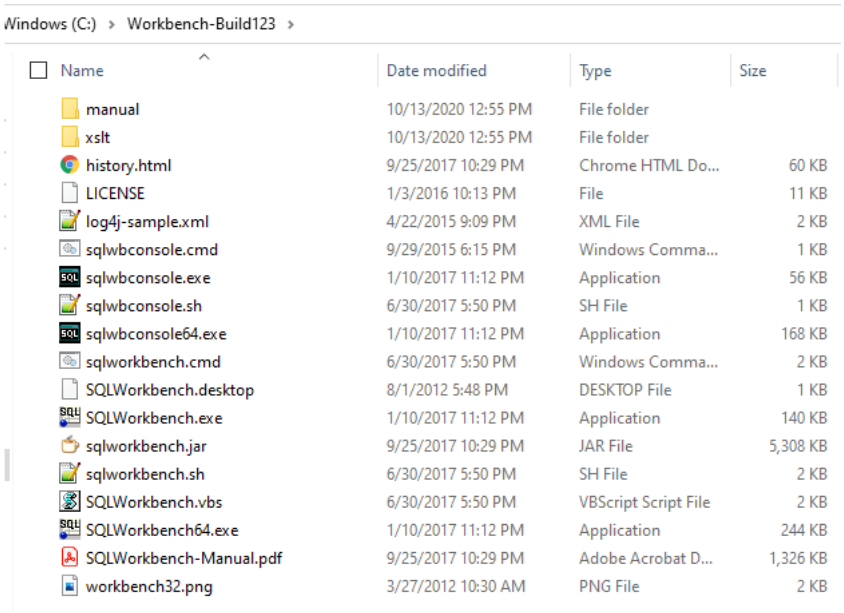
The following code contains sample commands for install and upgrade options:

- `msiexec /i <installer directory> TARGETDIR=<client directory> /quiet` for a fresh install. A fresh install creates the files but does not replace the files if they exist.
  For example: `msiexec /i InfaInstaller.msi TARGETDIR=C:\DDMClient /quiet`
- `msiexec /fa <installer directory> TARGETDIR=<client directory> /quiet` for an upgrade or repair. An upgrade or repair deletes old files and copies the files from the installer.
  For example: `msiexec /fa InfaInstaller.msi TARGETDIR=C:\DDMClient /quiet`

Create an installer that performs the following tasks during a fresh install:

1. Copies the `GenericJDBC.jar` file to the target directory: `<TARGETDIR>\JDBC\GenericJDBC.jar` `<TARGETDIR>` is the SQLWorkbench install directory.

   The following image shows the original SQLWorkbench folder structure:



2. Creates the following files in the specified location:

   `<TARGETDIR>\SQLWorkbench64_ddm.exe`

   `<TARGETDIR>\SQLWorkbench64_ddm.ini`

During an upgrade, the installer overwrites and updates the files.

Create and Package Required Files

To ensure that the installer can create or update the files, create and package the required files with the installer.

The installer requires the following files:

`GenericJDBC.jar`

The Dynamic Data Masking installation contains the JAR file that you must save on the client machine. The JAR file contains a Java agent, a transformer that intercepts the method calls of JDBC objects, and proxy classes for JDBC classes.

You can find the .jar file in the following location: `<Dynamic Data Masking installation directory>\Wrappers\jdbc\GenericJDBC.jar`

`SQLWorkbench64_ddm.exe`

Create a copy of the `SQLworkbench64.exe` file and name the file `SQLWorkbench64_ddm.exe`.

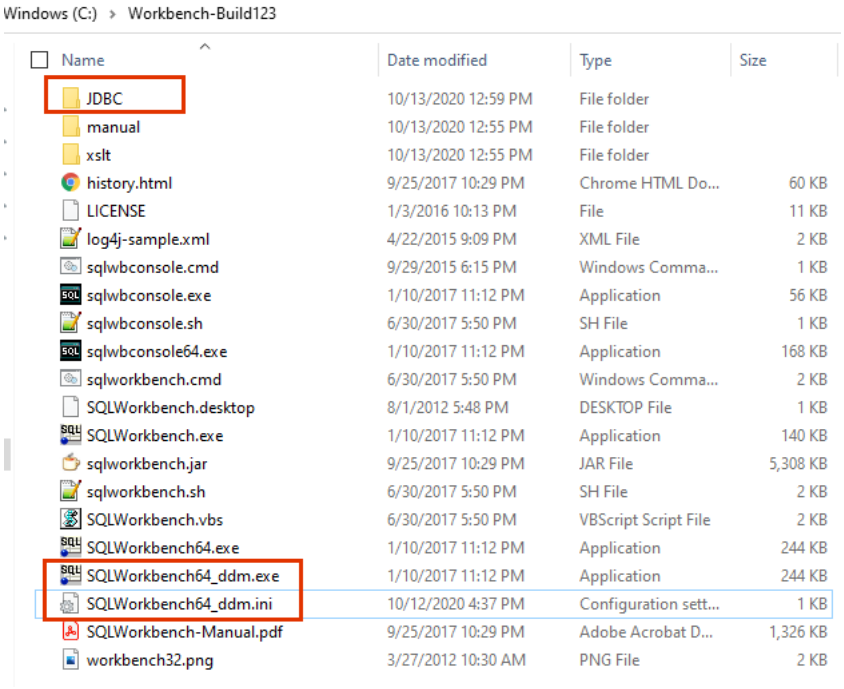
`SQLWorkbench64_ddm.ini`

Create an `SQLWorkbench64_ddm.ini` file with the following content:

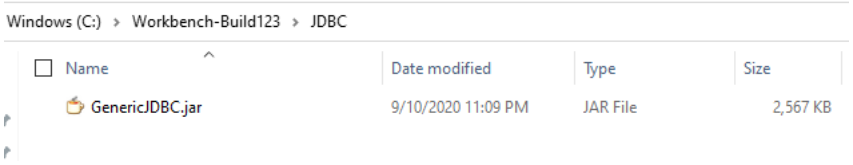
```
working.directory=.
classpath.1=sqlworkbench.jar
main.class=workbench.WbStarter
log.file.and.console=true
```

```
log.level=warning
vmarg.1= -javaagent:..\JDBC\GenericJDBC.jar=host:<host name>,port:<port number>
vmarg.2= -DDDM_GENJDBC_LOG=..\JdbcLogConfig.properties
```

The following image shows the SQLWorkbench folder structure after the installer deploys:



The following image shows the contents of the JDBC folder:

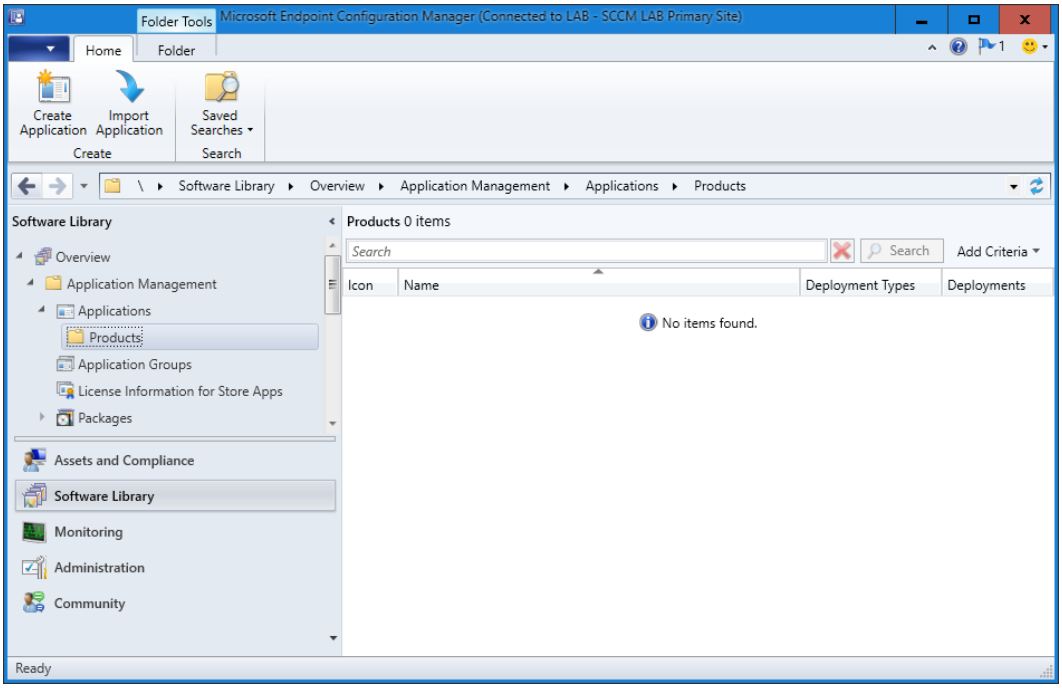


Note: To work with a connection in Dynamic Data Masking, use the `SQLWorkbench64_ddm.exe` file to open the client. To work directly with SQLWorkbench, use the `SQLWorkbench64.exe` file to open the client.

Using Microsoft Endpoint Configuration Manager to Deploy the Installer

1. Open Microsoft Endpoint Configuration Manager and navigate to **Software Library** > **Overview** > **Application Management** > **Applications**.

   You can choose to create and use a folder in the `Applications` folder.
   The following image shows a `Products` folder created in the `Applications` folder:

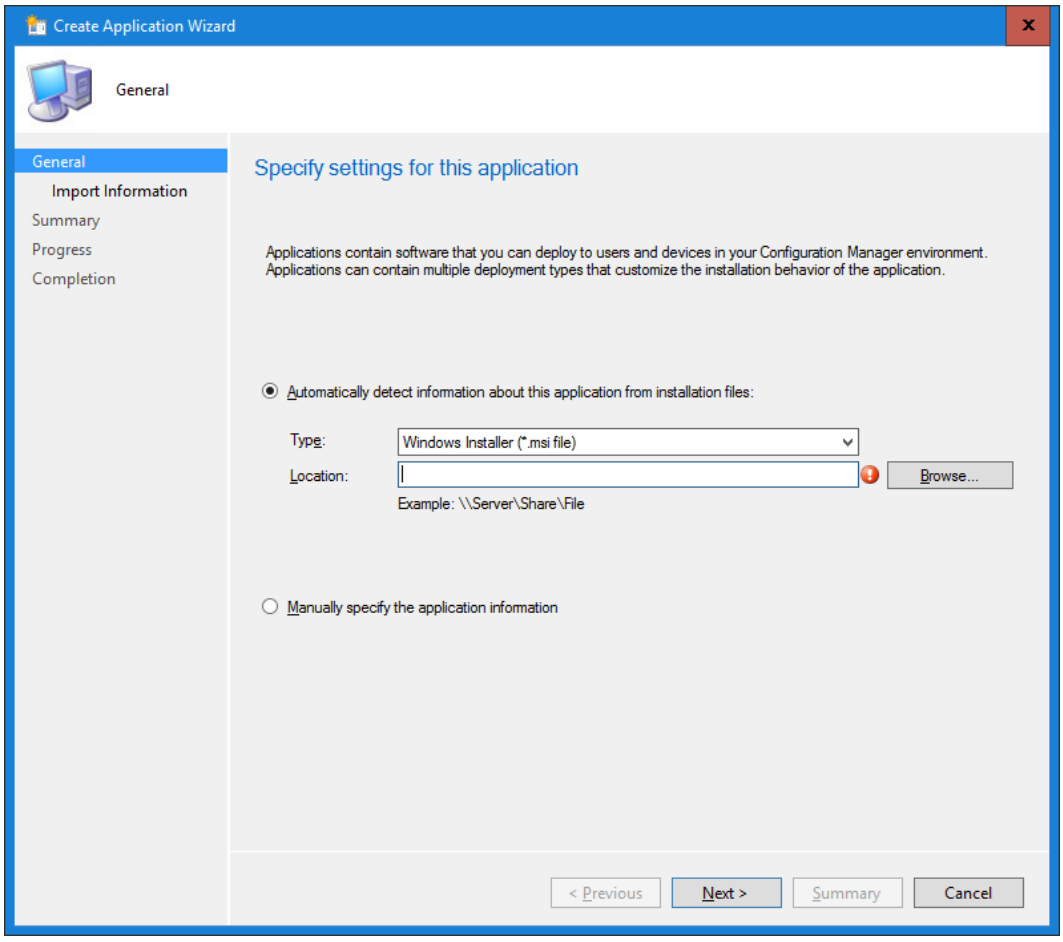


2. Click **Create Application**.

   The **Create Application Wizard** opens.

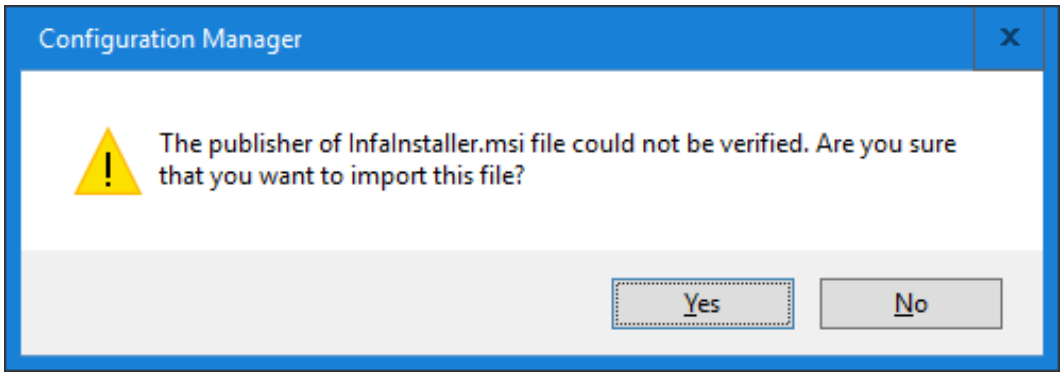
3. Choose to automatically detect information about the application from the installer files.

The following image shows the **General** page:



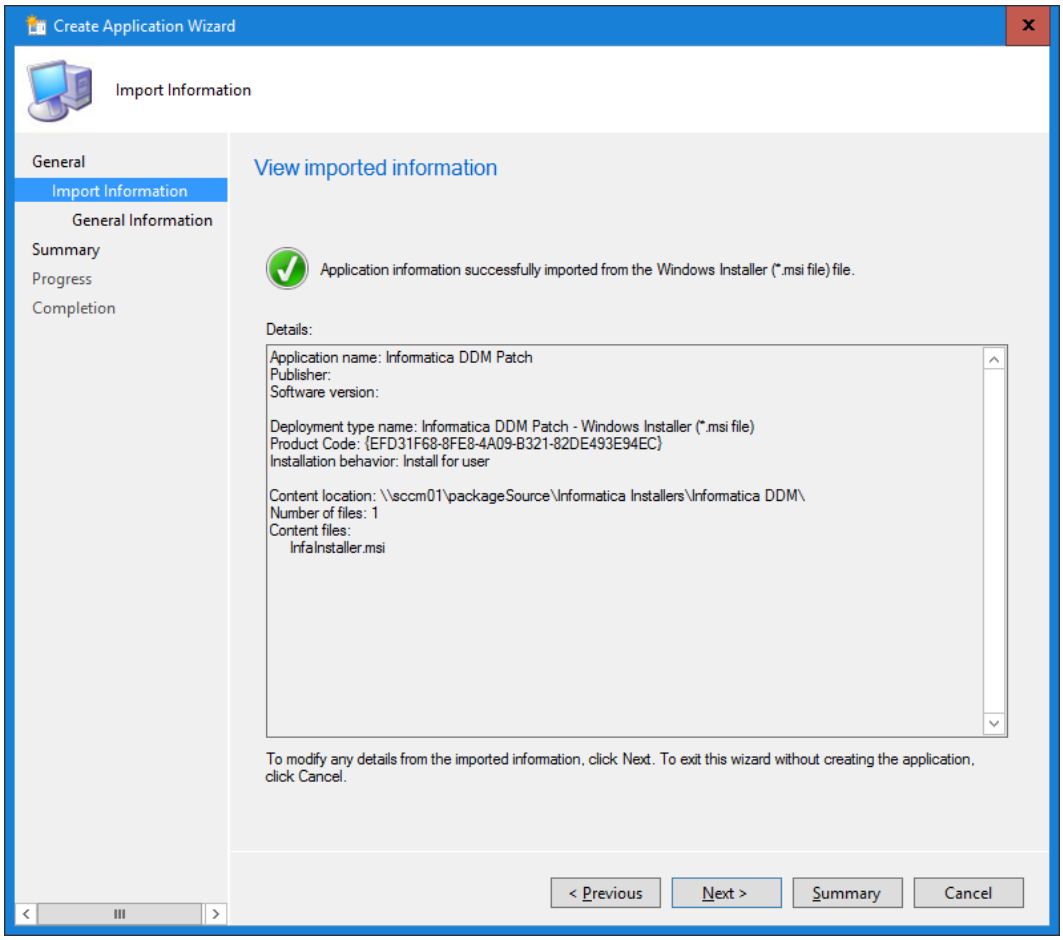
4. Click **Browse** and choose the installer package location.
5. Click **Next**.
6. If prompted to confirm the import, click **Yes** to confirm.

   The following image shows the confirmation prompt window:



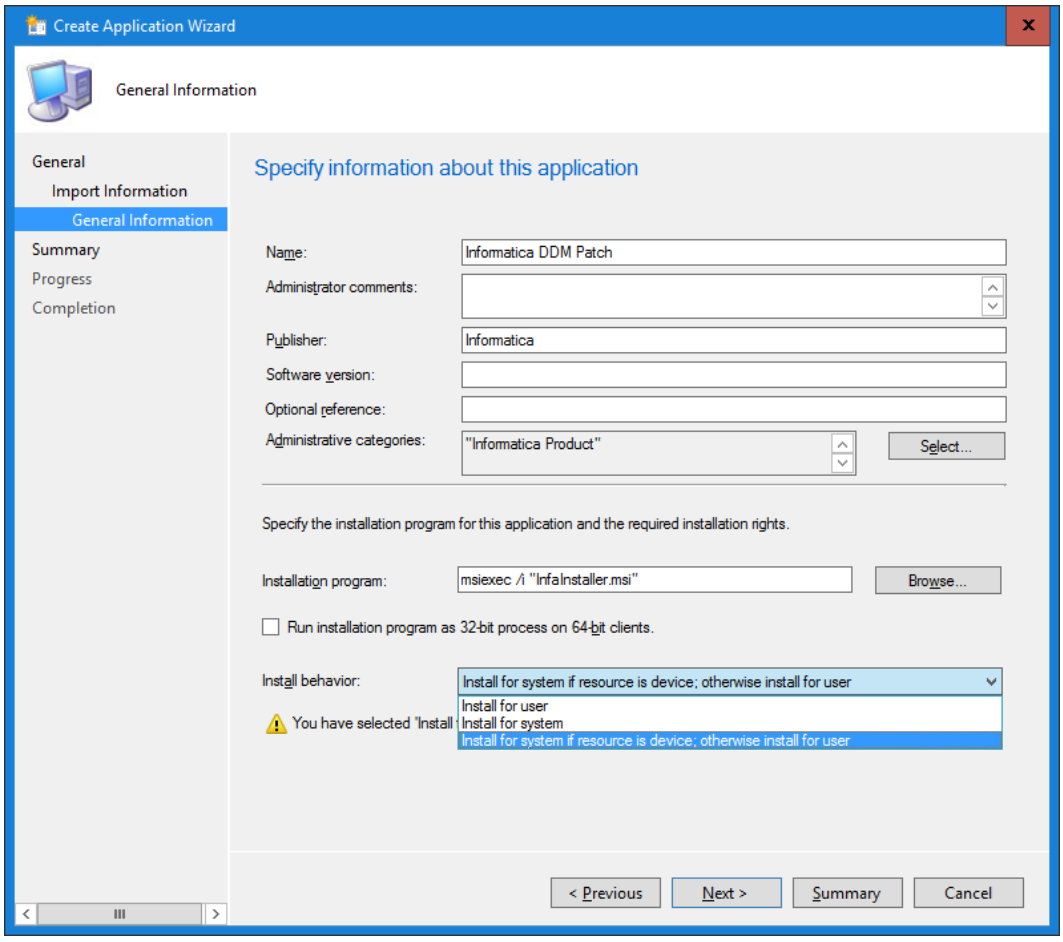
   The **Import Information** page opens.

The following image shows the **Import Information** page:



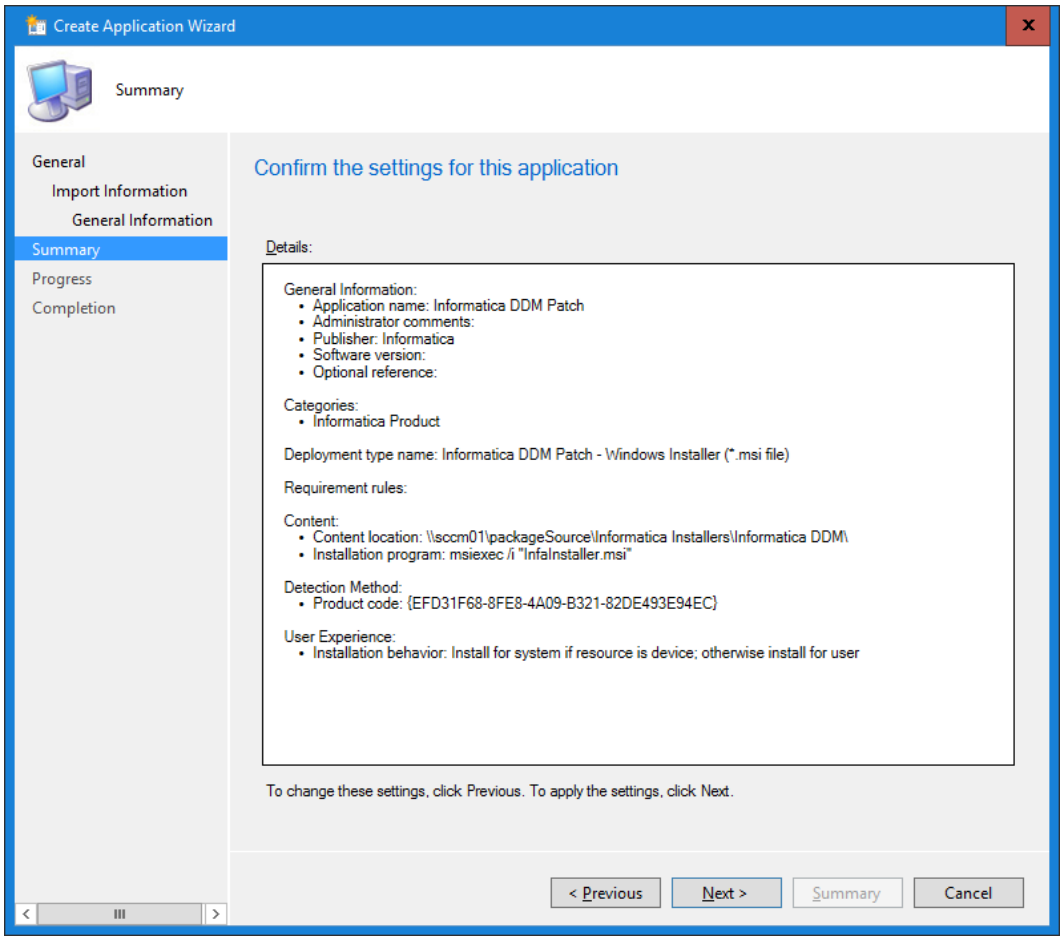
7. Click **Next**.
8. On the **General Information** page, enter the required fields.
9. Enter a name for the application and information about the application as required.
10. From the **Install Behavior** list, choose **Install for system if resource is device: otherwise install for user** option and click **Next**.

The following image shows information entered on the **General Information** page:



11. To confirm the settings, click **Next** on the **Summary** page.

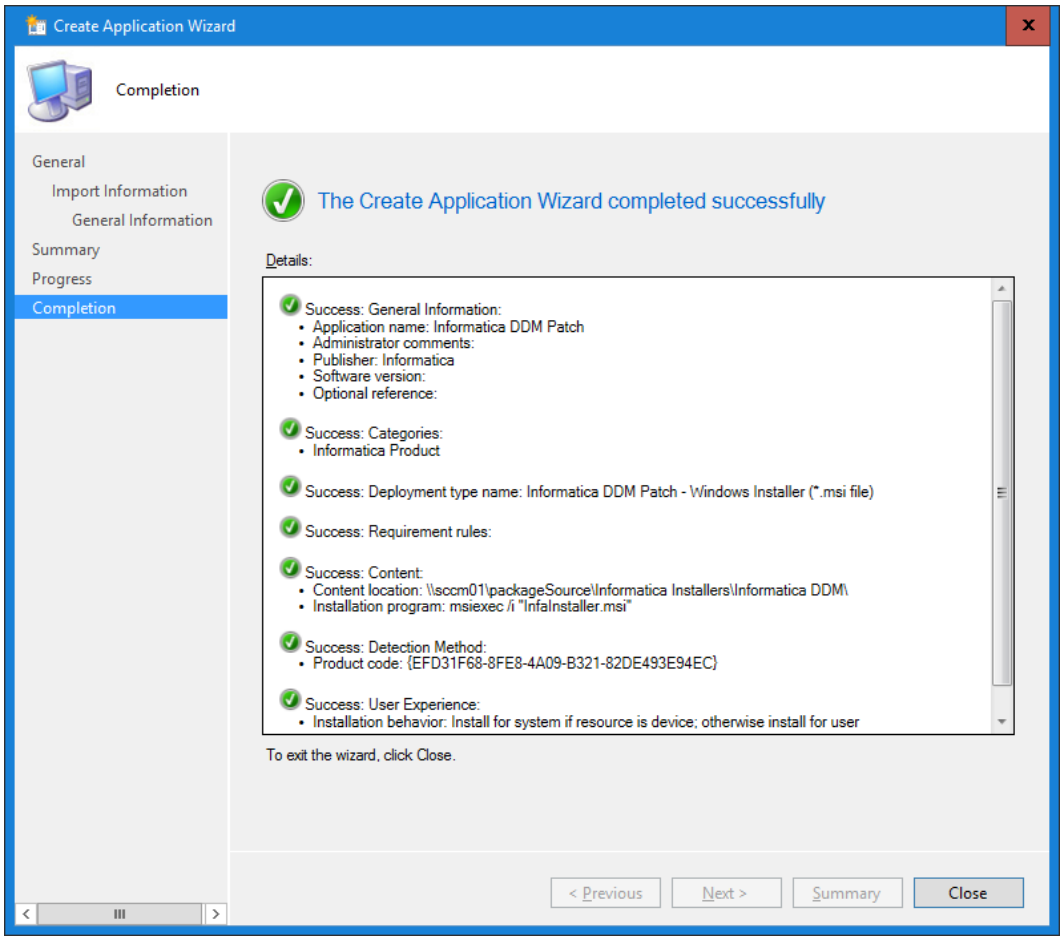
The following image shows information on the **Summary** page:



The application creation begins and when it finishes, the **Completion** page appears.

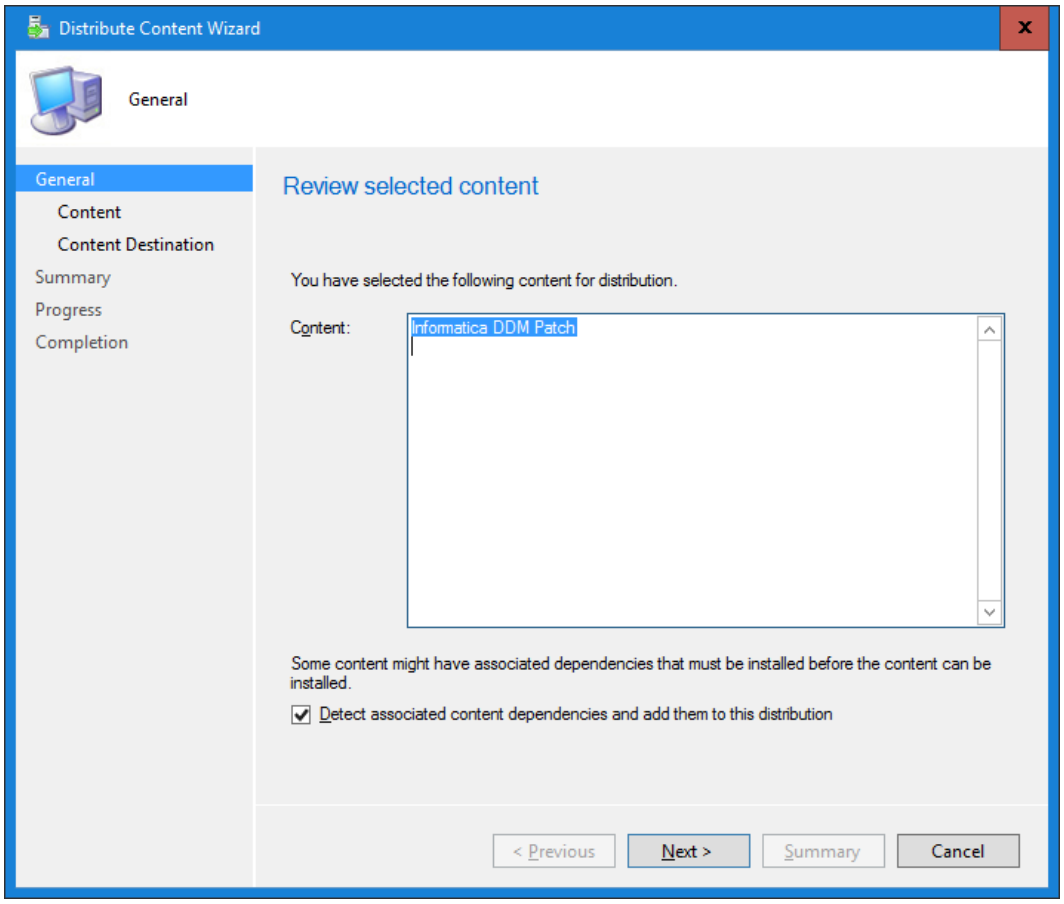
12. Click **Close** to close the wizard and return to the main page.

The following image shows the **Completion** page:



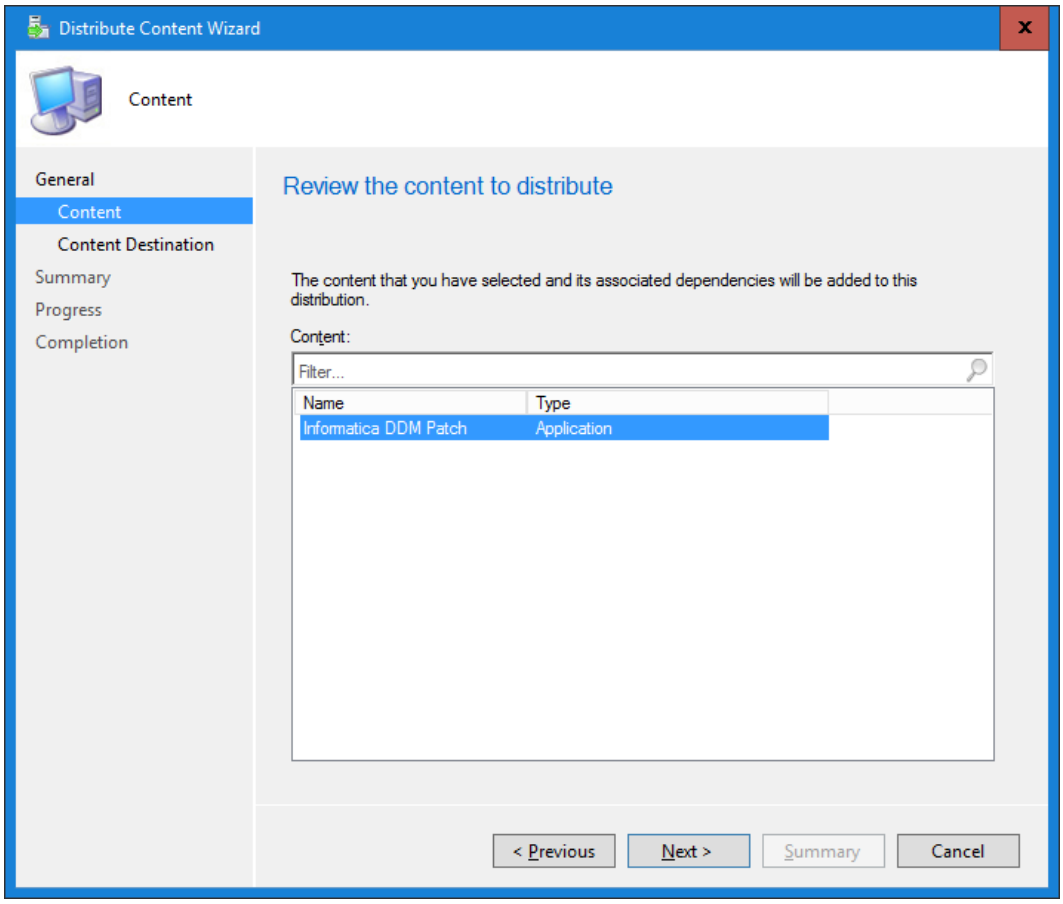
13. Click **Distribute Content** to open the Distribute Content Wizard.
14. On the **General** page, verify the information and click **Next**.

The following image shows the **General** page:



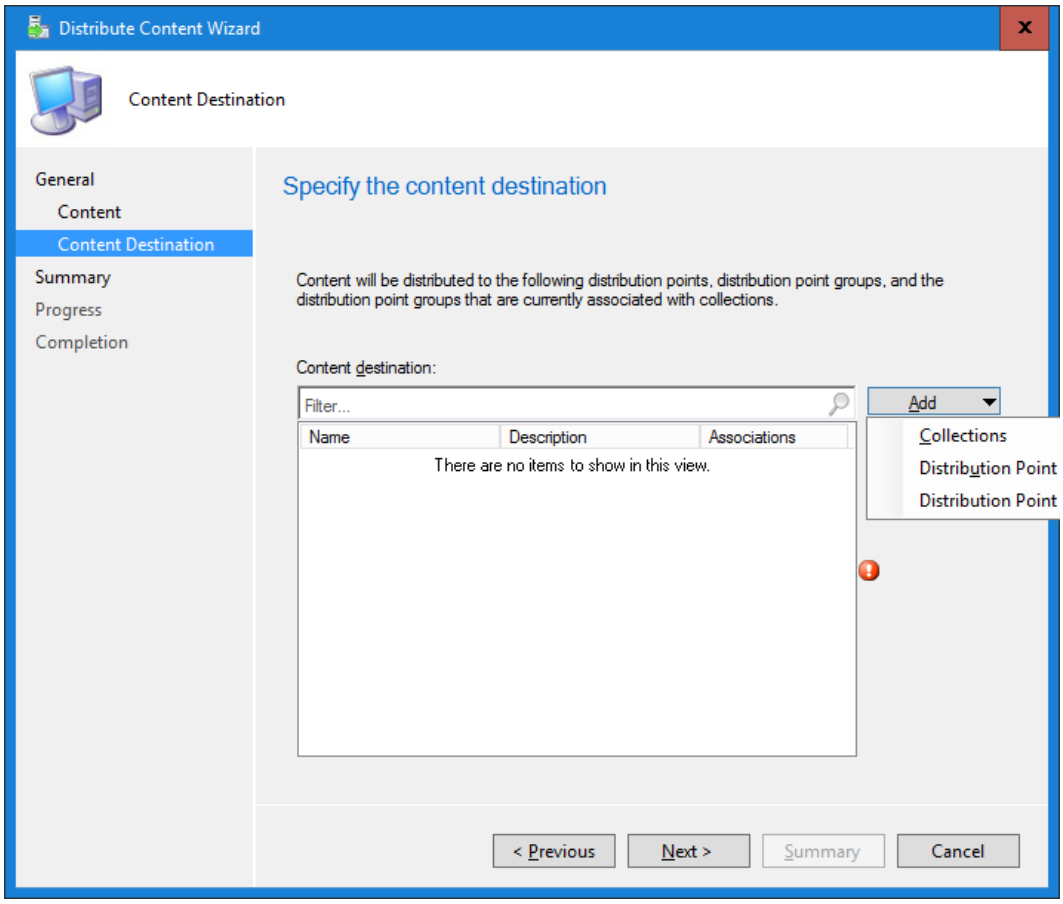
15. On the **Content** page, select the application and click **Next**.

The following image shows the **Content** page:

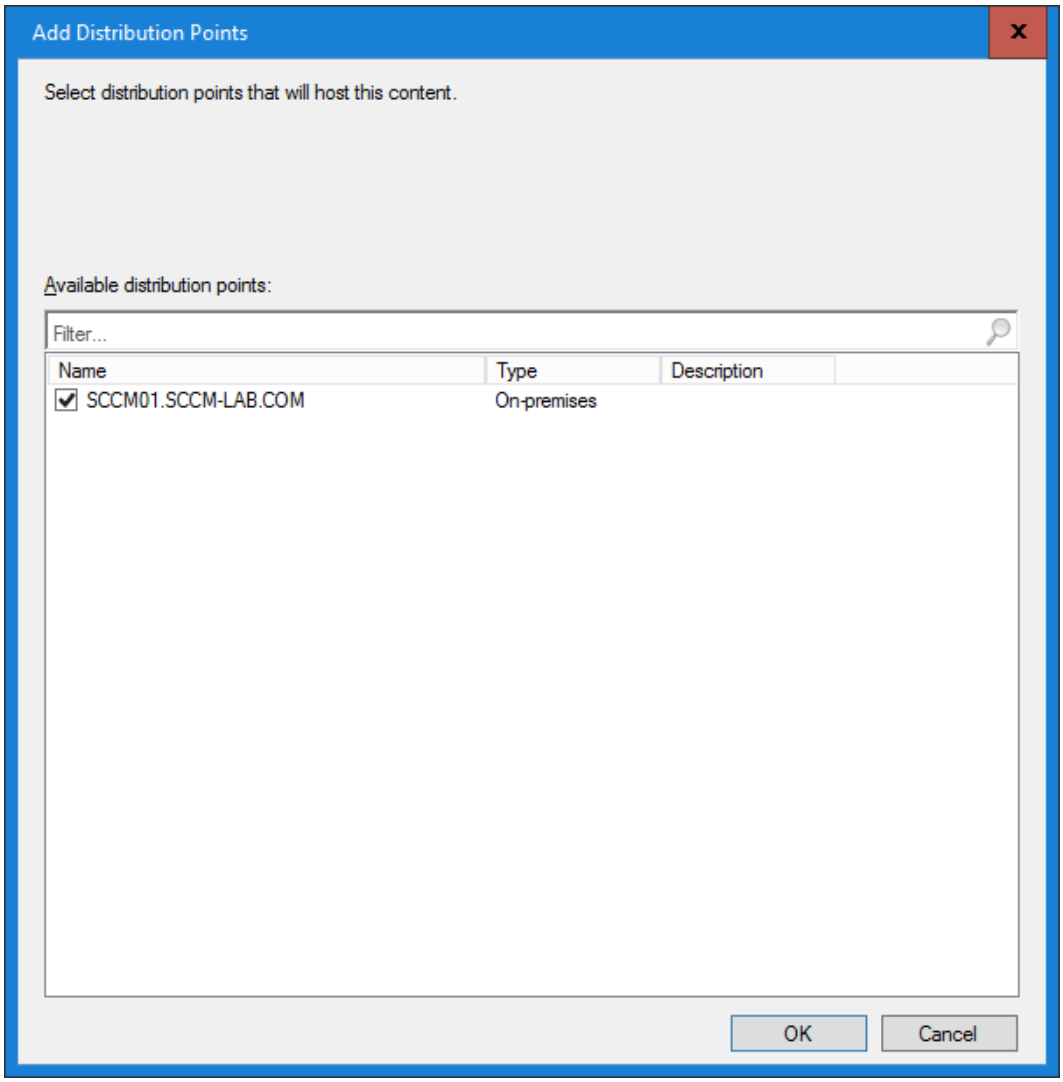


16. On the **Content Destination** page, click **Add Distribution Points**. On the **Add Distribution Points** dialog box that opens, select the required distribution points. Click **OK** to close the dialog box and then click **Next**.

The following image shows the **Content Destination** page:

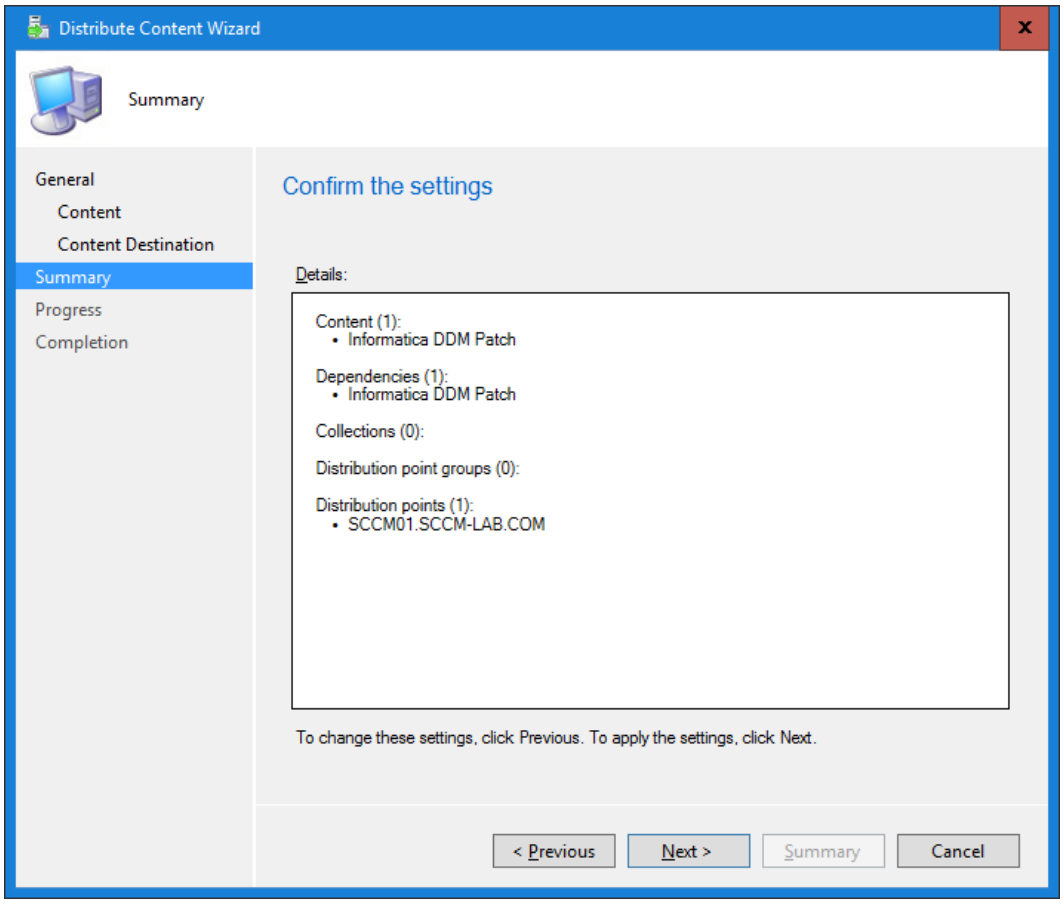


The following image shows the **Add Distribution Points** dialog box:



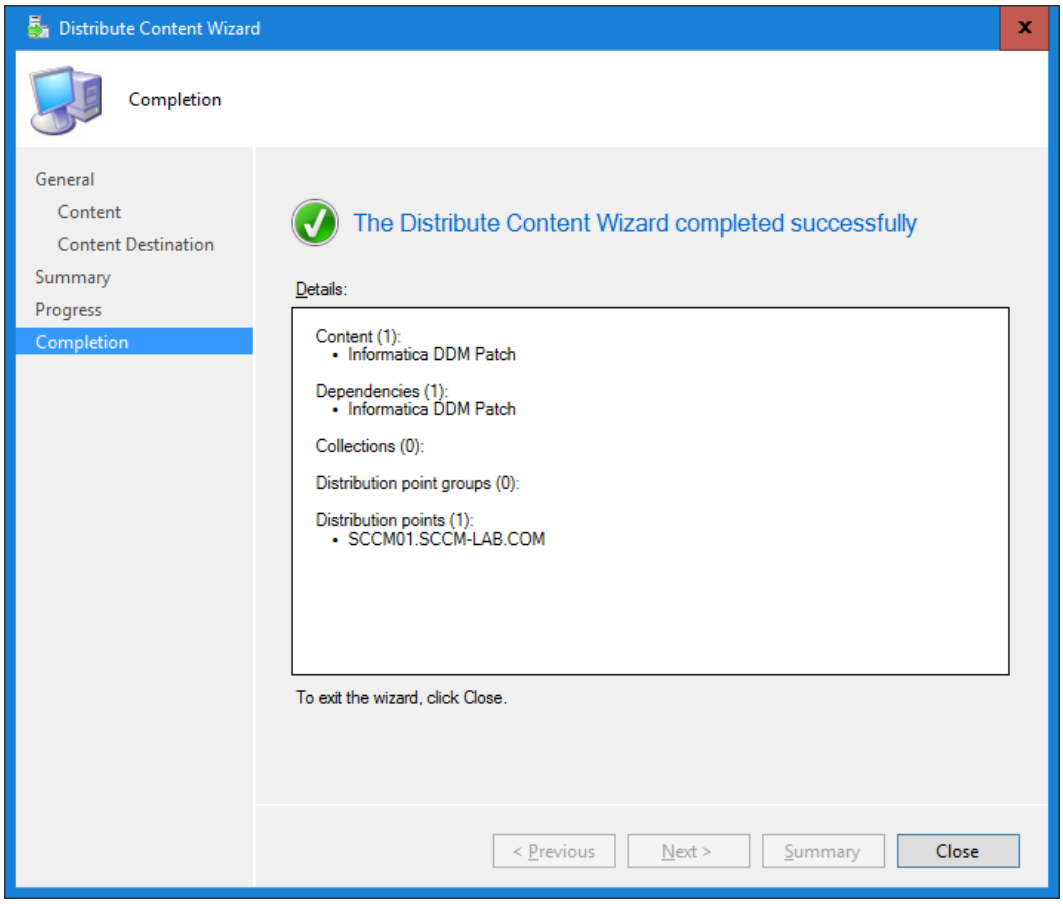
17. Review the information on the **Summary** page and click **Next**.

The following image shows the **Summary** page:



18. Click **Close** to close the wizard and return to the main page.

The following image shows the **Completion** page:



19. Click **Deploy**.

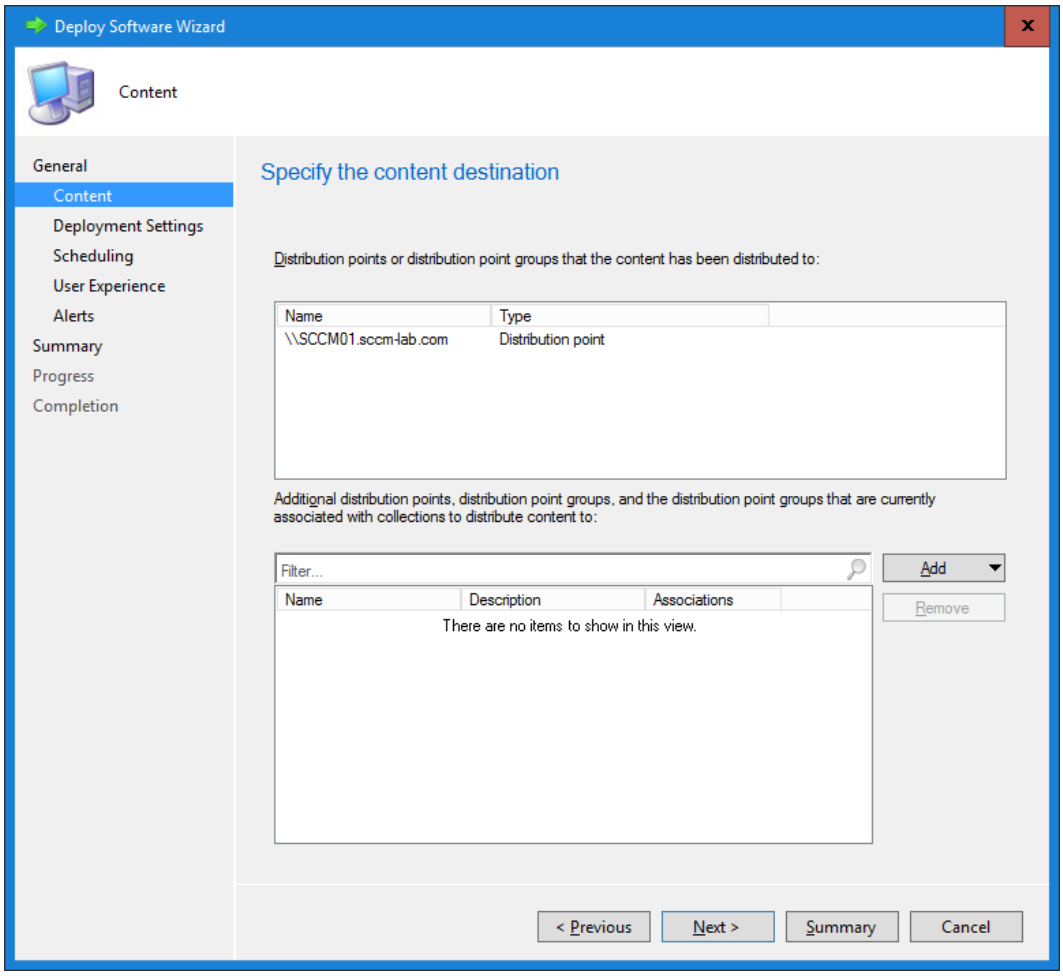
    The Deploy Software Wizard opens.

20. On the **General** tab, click **Browse** to select the Collection.

    Choose the required user or system. Click **OK** on the **Select Collection** window and then click **Next**.

21. On the **Content** page, click **Next**.

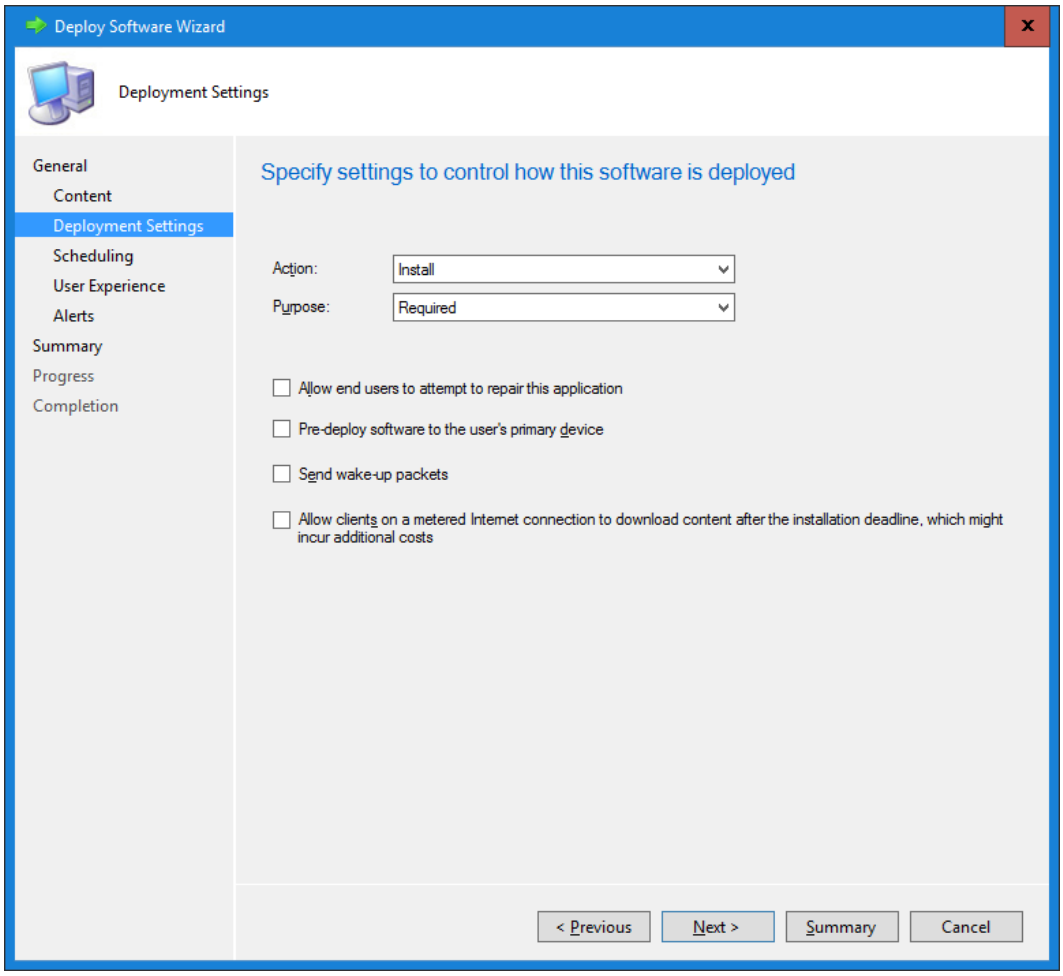
The following image shows the **Content** page:



22. On the **Deployment Settings** page, select the **Action** as Install and the **Purpose** as Required or Available and then click **Next**.

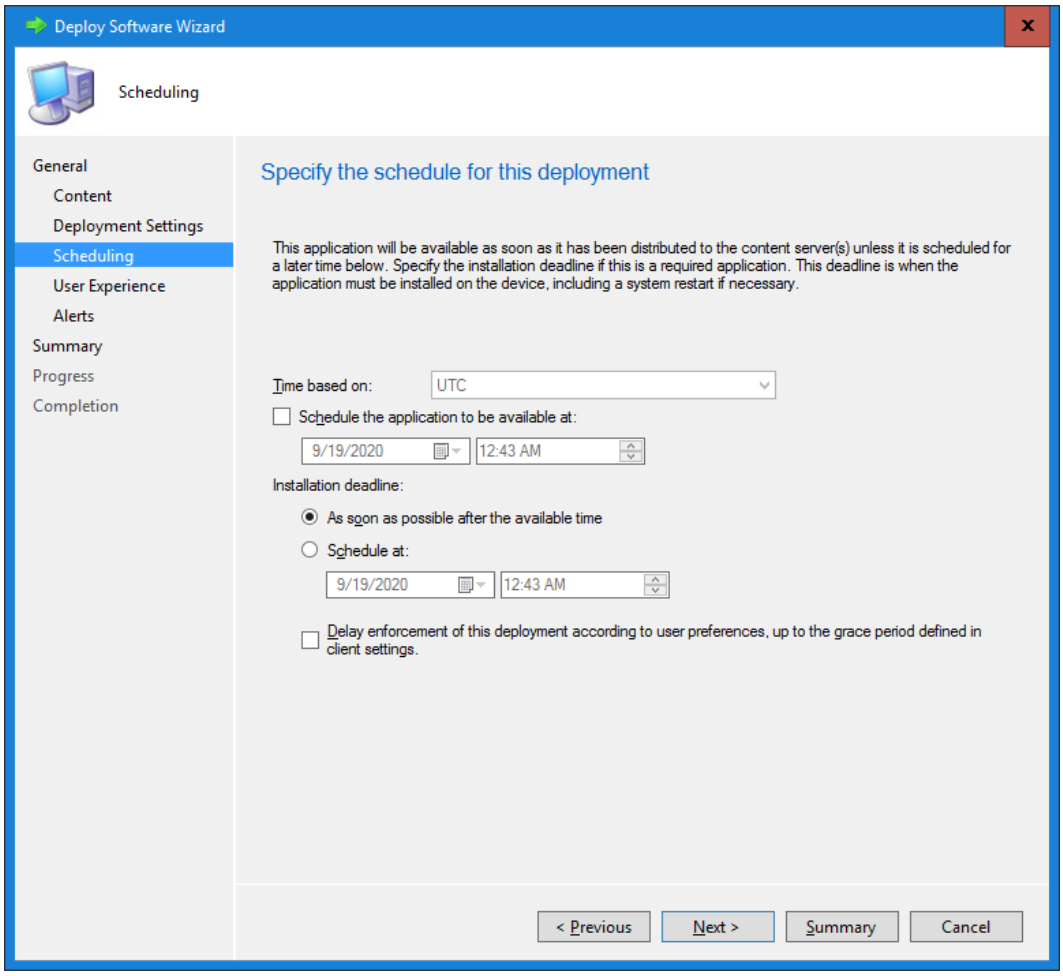
    Required indicates a forced installation and Available indicates an on-demand installation.

The following image shows the **Deployment Settings** page:



23. On the **Scheduling** page, select the date on which to make the application available. If you chose a forced installation, enter a deadline. Click **Next**.

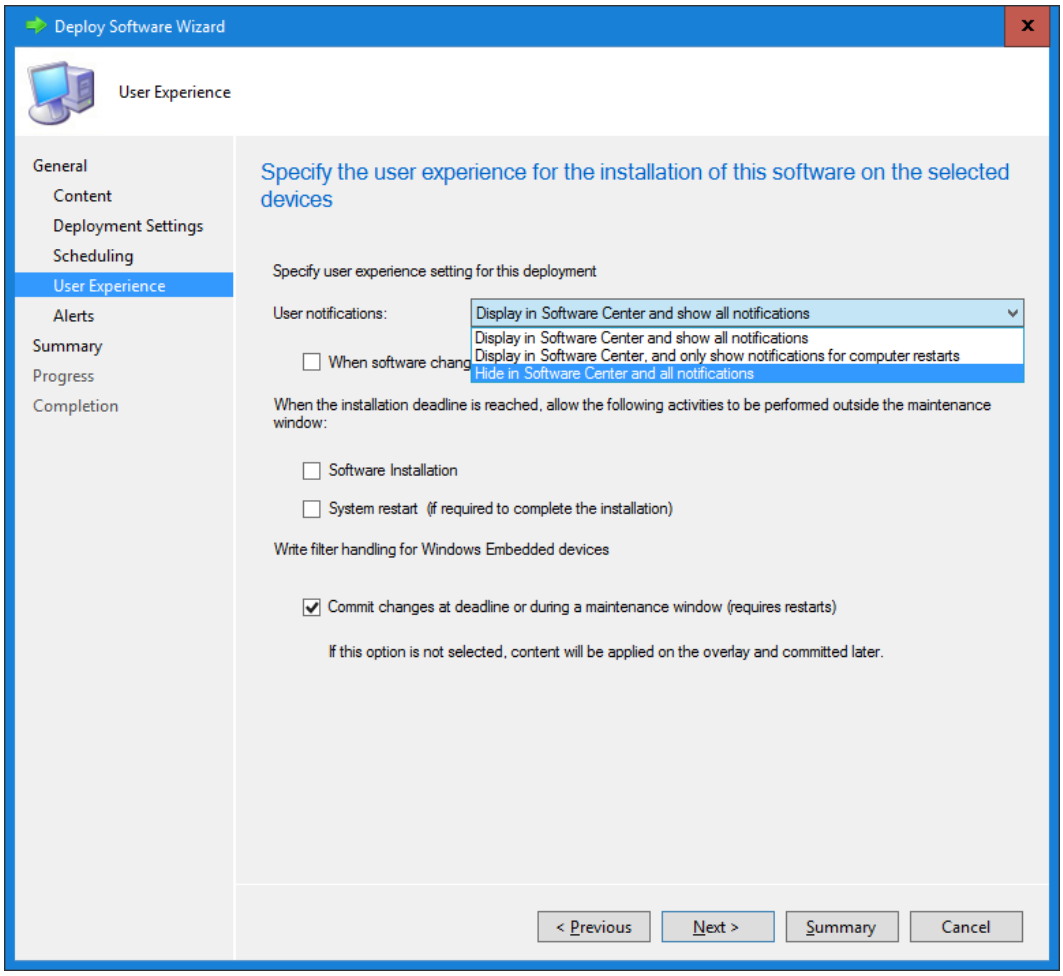
The following image shows the **Scheduling** page:



24. On the **User Experience** page, configure the user notification information and then click **Next**.

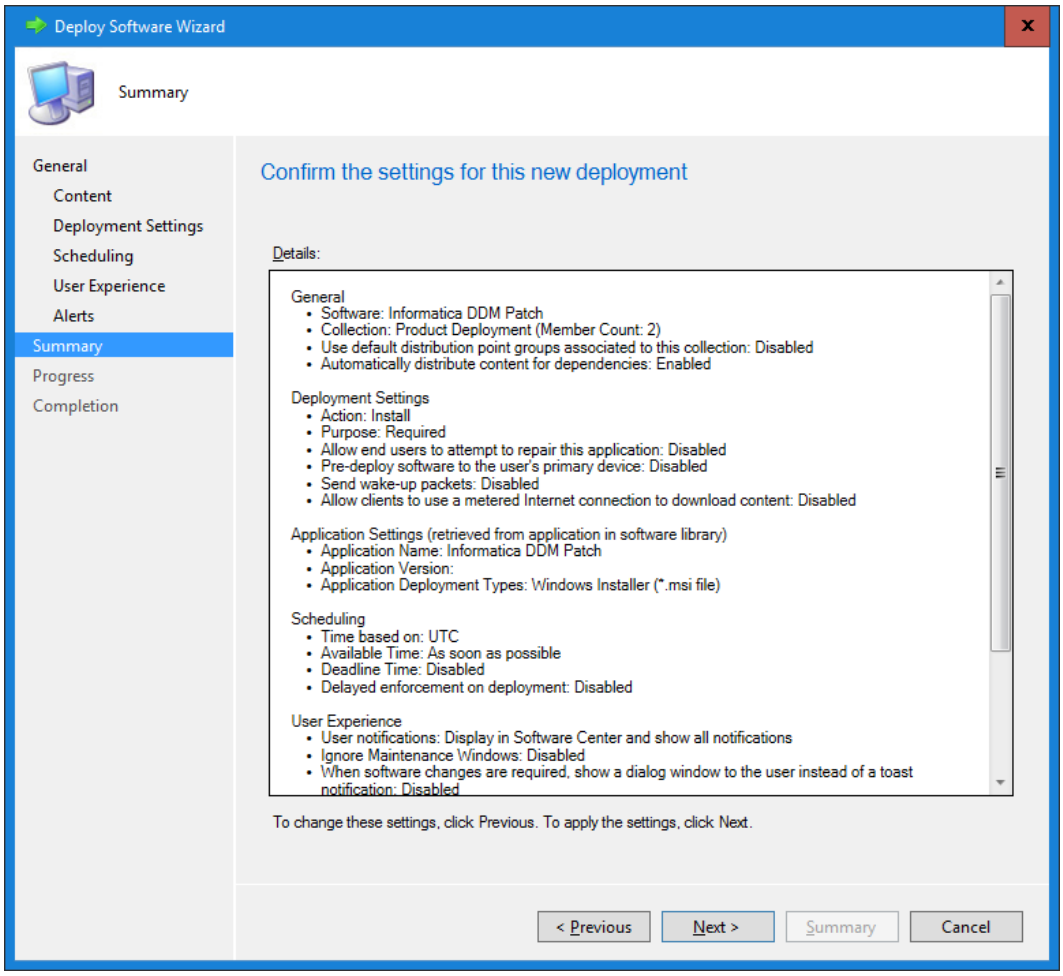
    For silent and unattended installations, select **Hide in Software Center and all notifications**. For on-demand installations, select **Display in Software Center, and only show notifications for computer restarts**.

The following image shows the **User Experience** page:



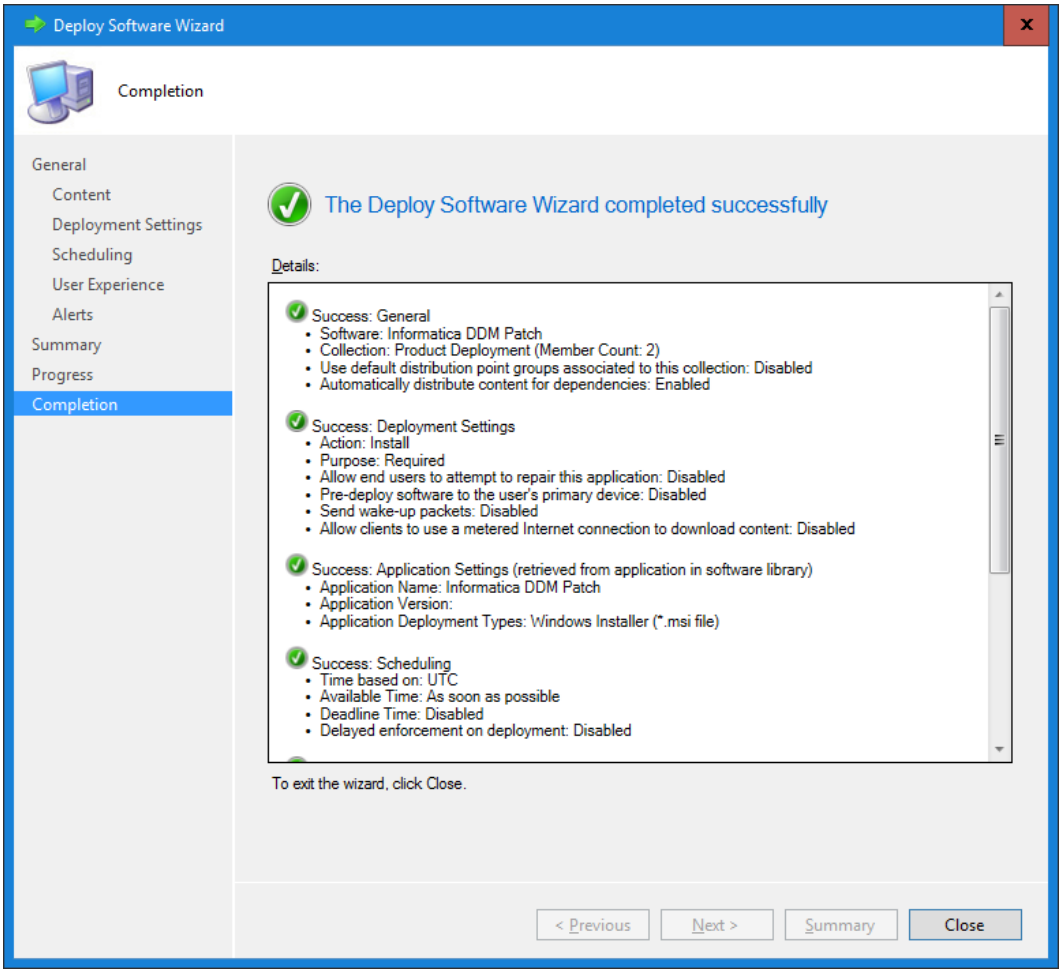
25. Configure Alerts information if required and click **Next**.
26. On the **Summary** page, click **Next**.

The following image shows the **Summary** page:



27. On the **Completion** page, review the settings and click **Close** to complete the deployment creation.

    The following image shows the **Completion** page:



    The application deploys or is made available from the Software Center at the next user or computer policy evaluation cycle.

See the Microsoft product documentation for detailed steps and instructions.

Using Other Deployment Management Tools

You can use other software management tools to deploy the installer package.

This section describes how to deploy the installer using the following tools:

- Microsoft Endpoint Manager
- Active Directory Group Policy Object

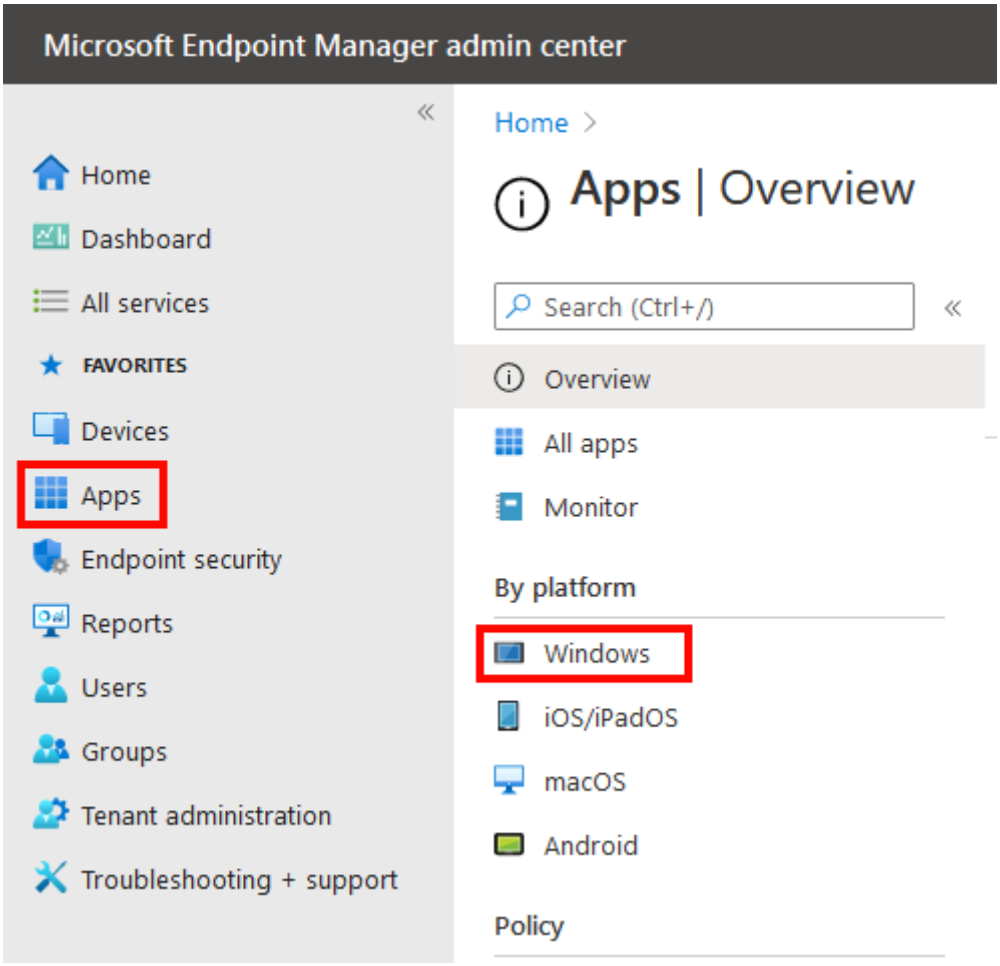
This is not a comprehensive list of tools. You can use any similar configuration deployment tool. Use the high-level steps as a reference if you work with other tools.

Using Microsoft Endpoint Manager to Deploy the Installer

1. Open the Microsoft Endpoint Manager console.

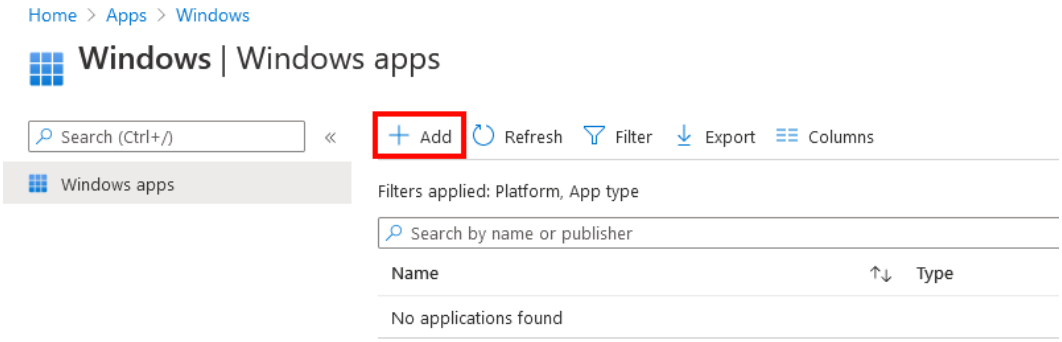
2. Click the **Apps** blade on the panel on the left and select **Windows**.

   The following image shows the **Apps** blade on the Microsoft Endpoint Manager:



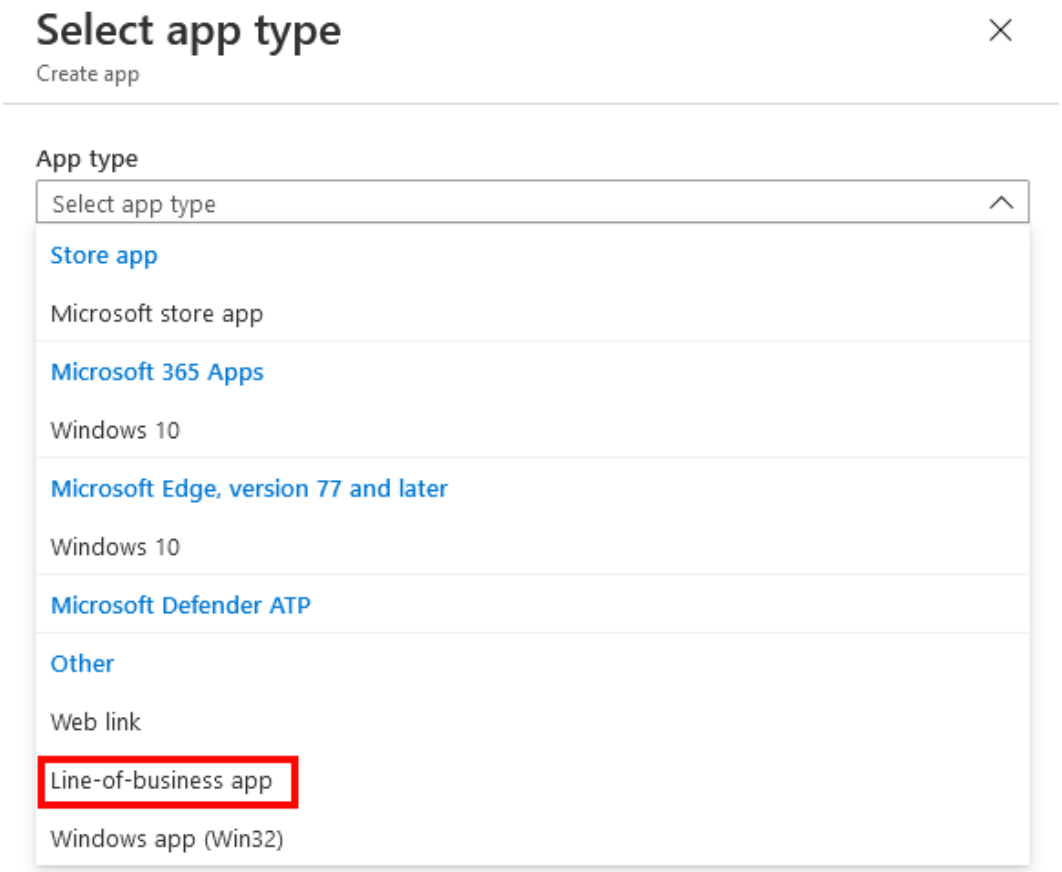
3. Click **Add**.

   The following image shows the **Windows apps** dialog box:



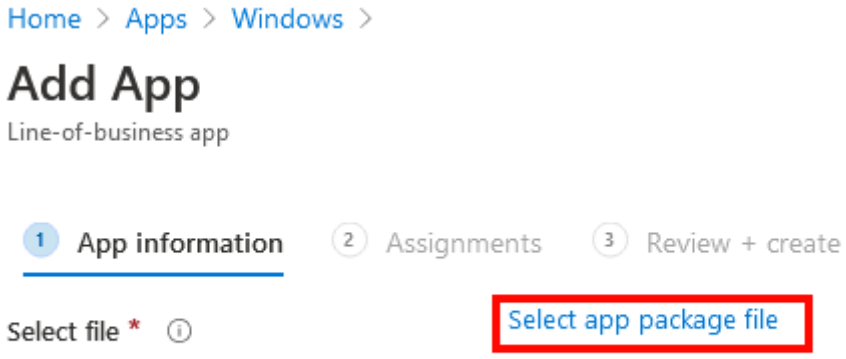
4. From the list of app types, choose **Line-of-business** and click **Select**.

The following image shows the list of app types in the **Windows apps** dialog box:



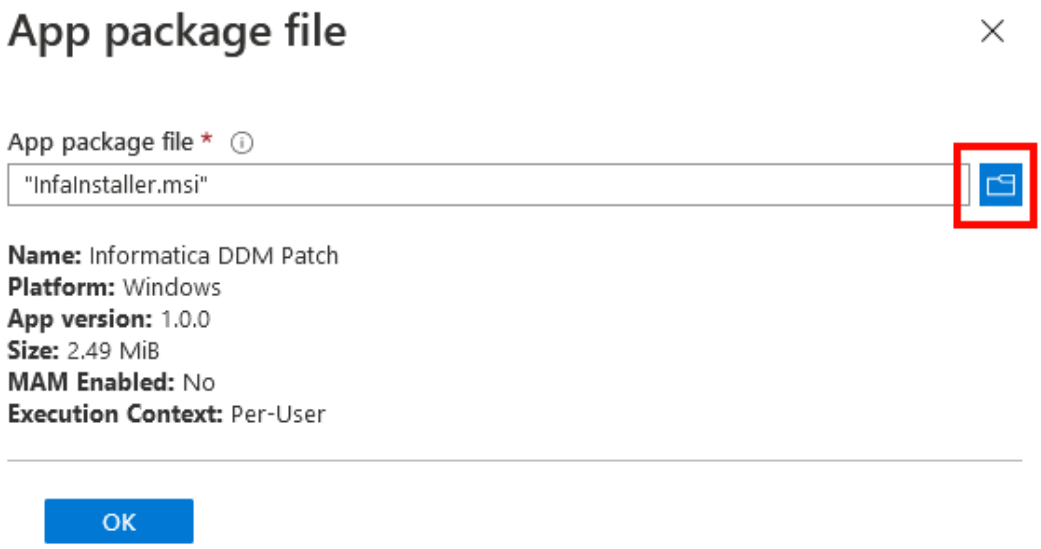
5. Click **Select app package file**.

   The following image shows the **App information** tab on the **Add App** dialog box:



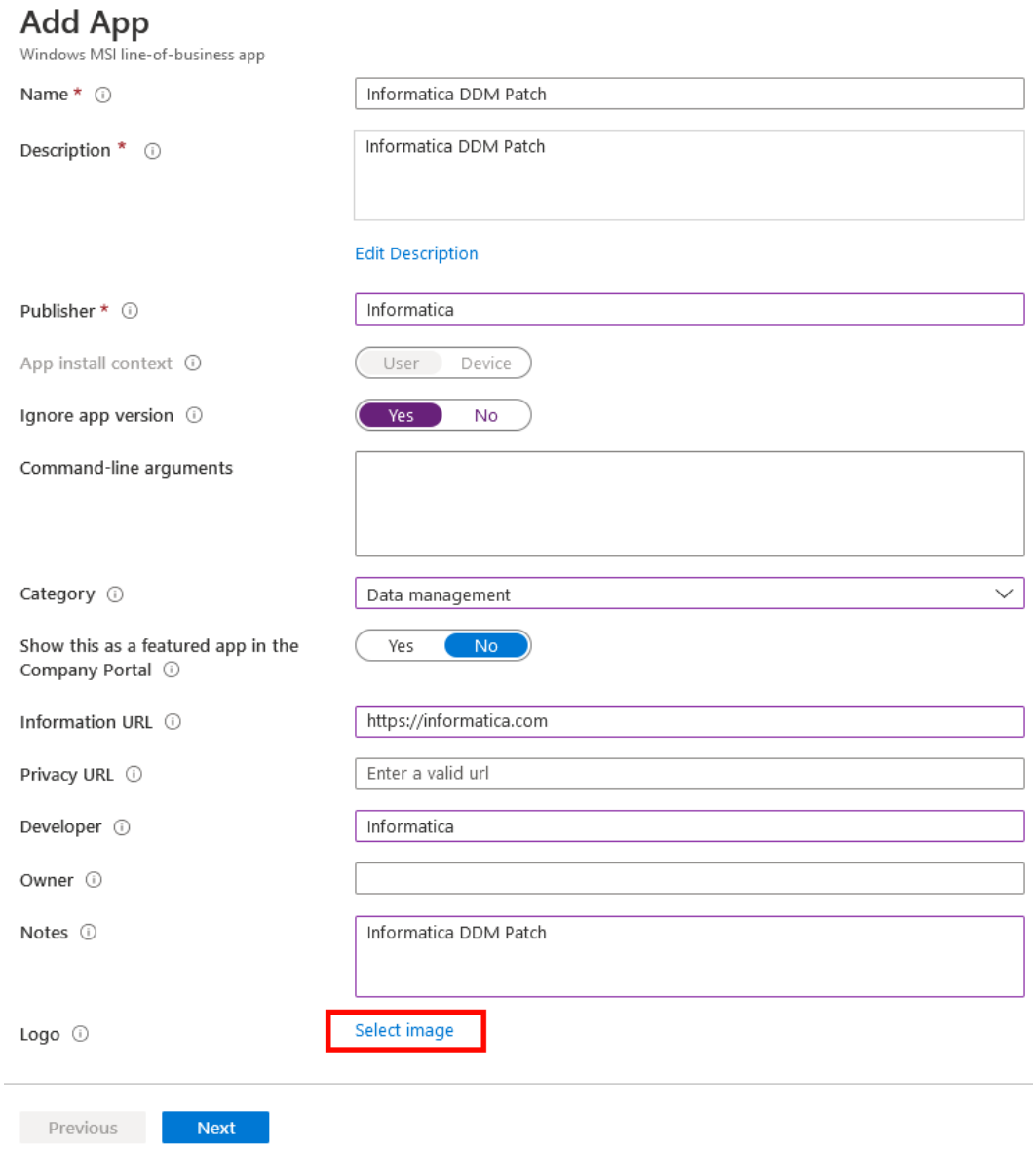
6. Click the **Browse** button and browse and select the required installer package. Click **Open** and then click **OK** to select the package.

The following image shows the **App package file** dialog box:



7. On the **Add App** window, update information if required. To add an application logo, click **Select image**, upload a custom image, and click **OK**.

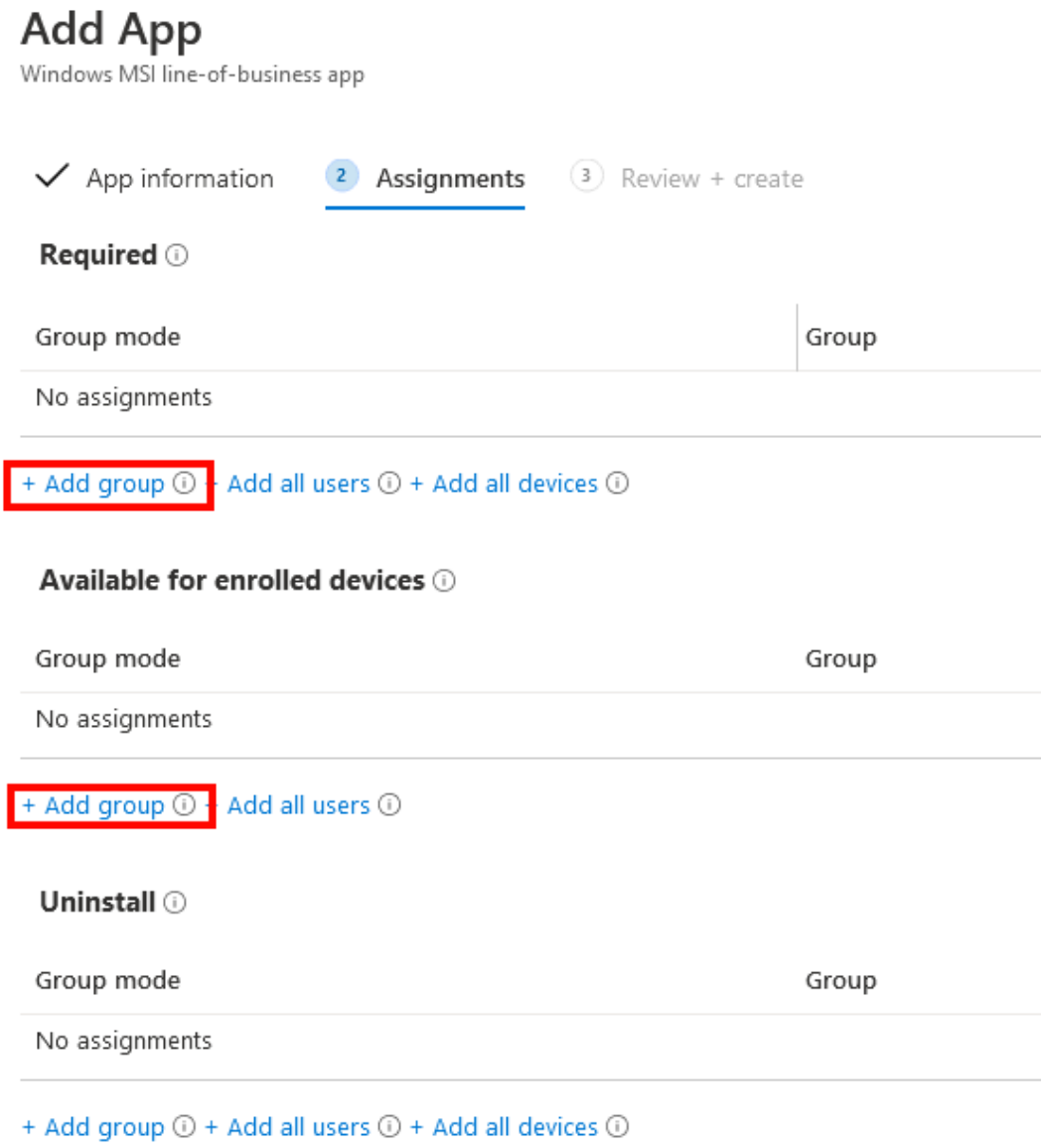
The following image shows the **Add App** dialog box:



8. Click **Next**.
9. On the **Assignments** tab, choose the type of deployment and click **Add group** to add a deployment.

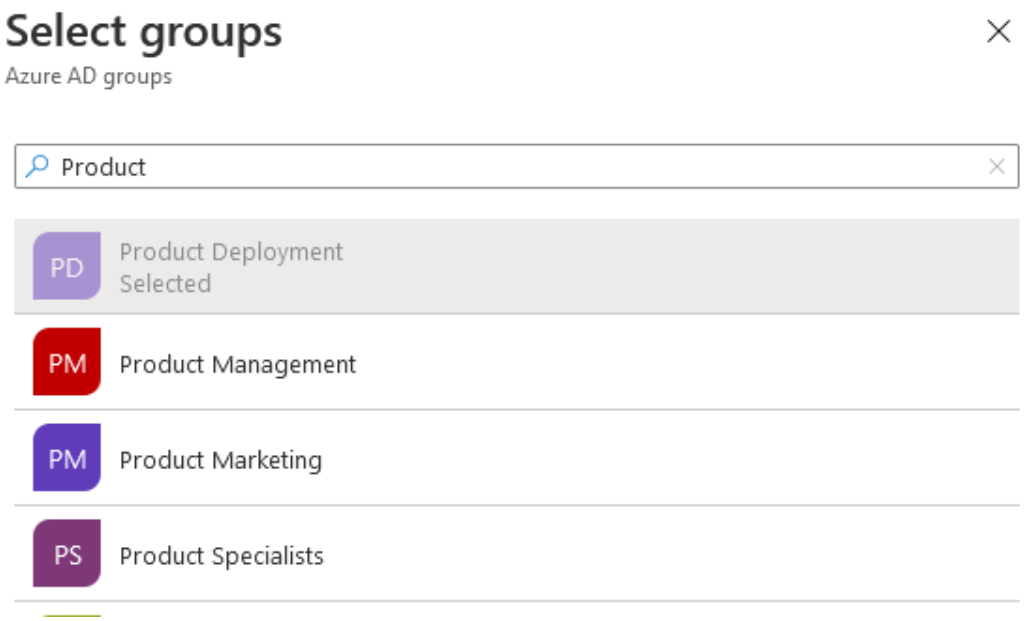
   Use **Required** for forced installations and **Available for enrolled devices** for on-demand installations from a company portal application.

The following image shows the **Assignments** tab:



10. Choose the required groups and click **Select**.

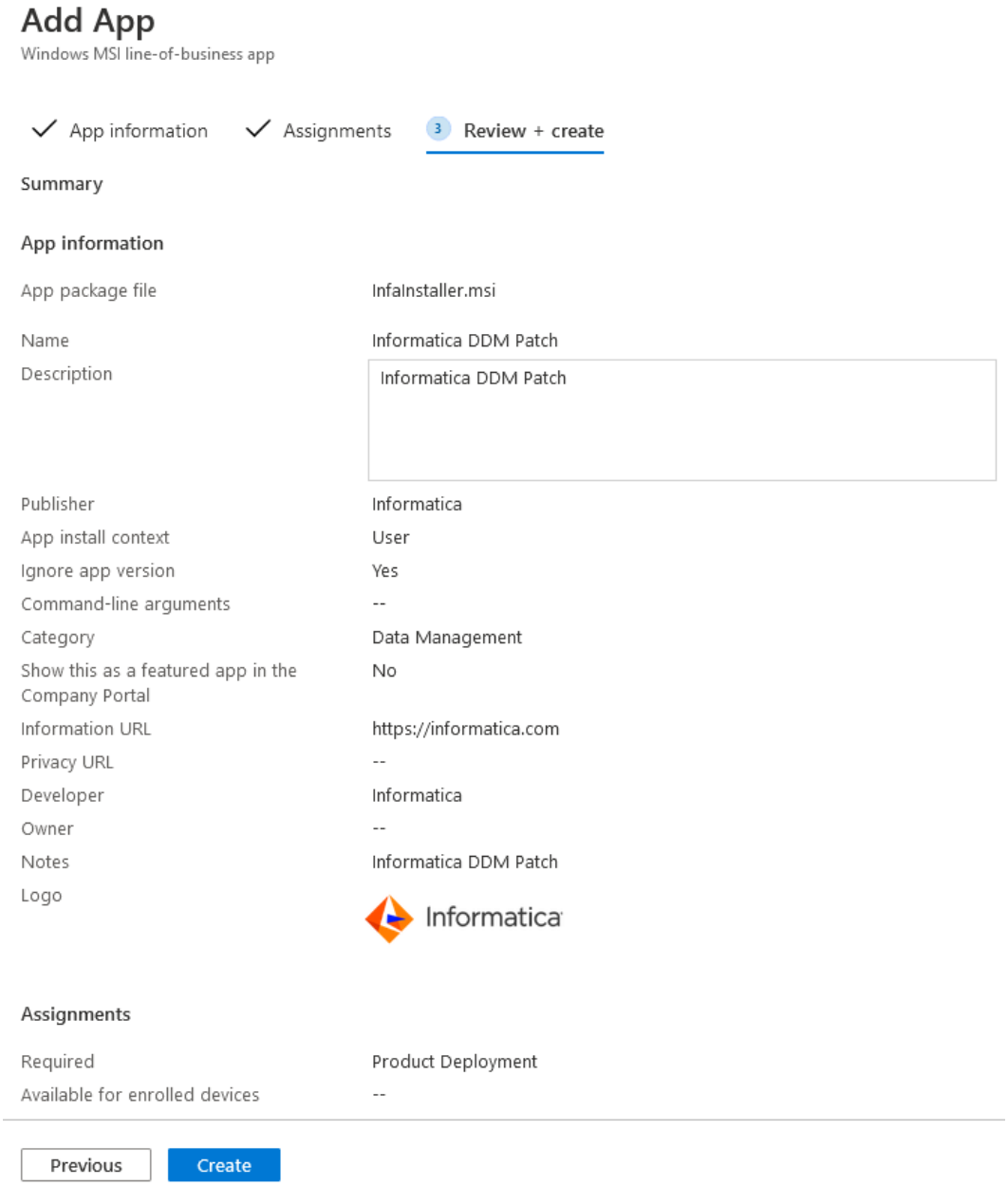
The following image shows the **Select groups** list:



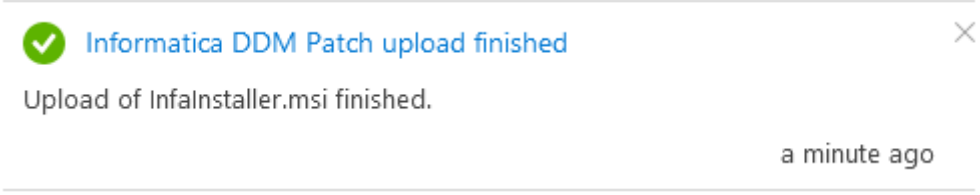
11. Click **Next**.

12. On the **Review+create** tab, review the summary details and click **Create**.

The following image shows the **Review+create** tab:



A status message indicates the status, and a successfully uploaded message appears when it finishes. The following image shows the message that appears when the application file upload finishes:



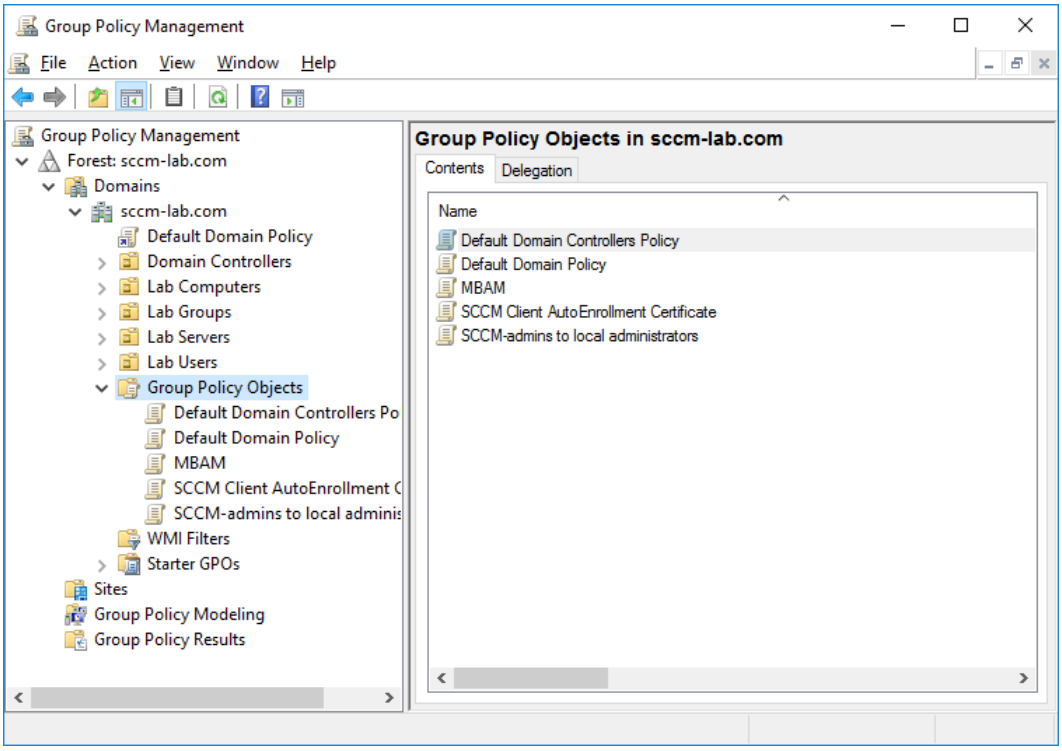
Replication of the application across regions takes some time after the file is successfully uploaded. For forced installations, the application deploys after it is replicated and the systems sync policies. For on-demand installations, the application deploys when it is listed on the company portal application.

See the Microsoft product documentation for detailed steps and instructions.

Using Active Directory Group Policy Object to Deploy the Installer

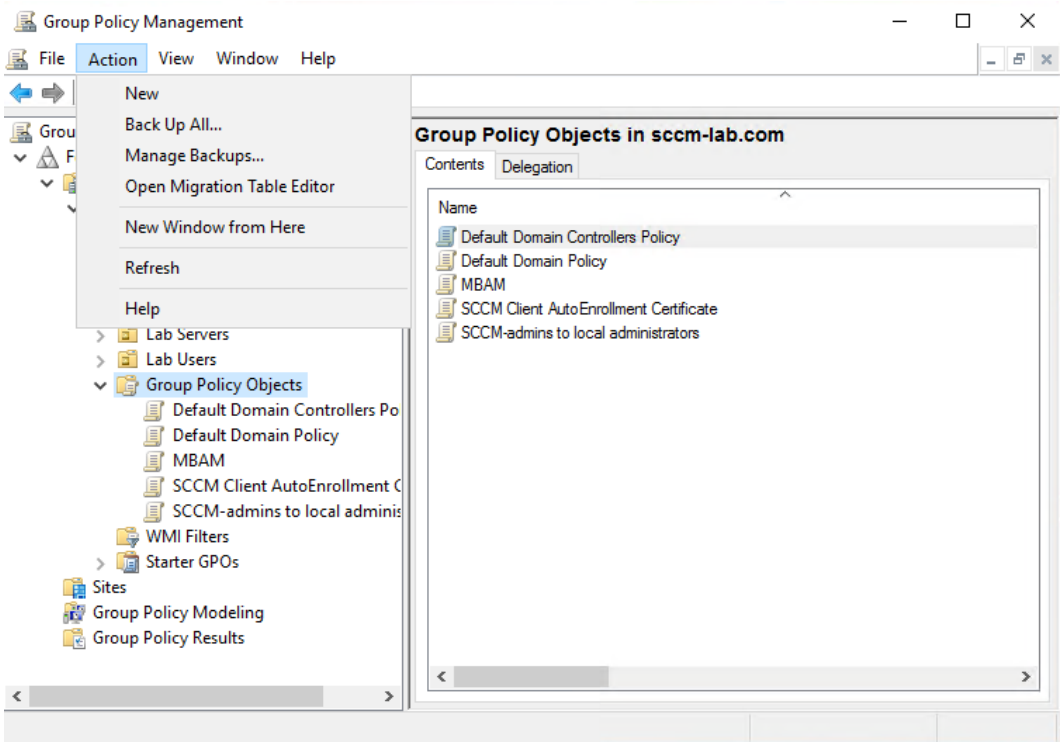
1. Open Group Policy Manager.
2. Navigate to **<Forest name>** > **Domains** > **<Domain name>** > **Group Policy Objects**.

   The following image shows the **Group Policy Management** window:

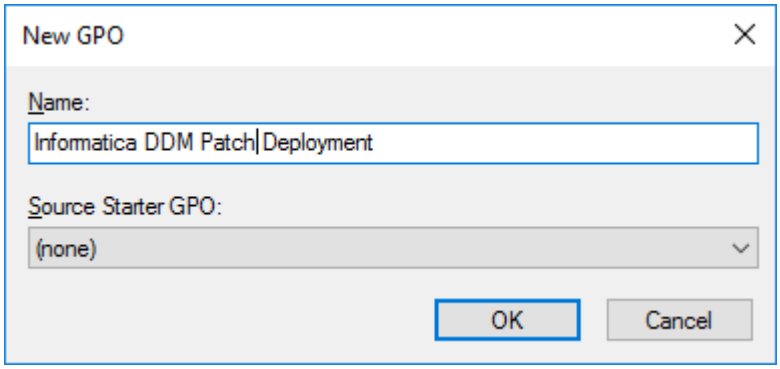


3. Click **Actions** > **New**. Enter a relevant name for the object and click **OK**.

The following image shows the **Actions** menu in the **Group Policy Management** window:

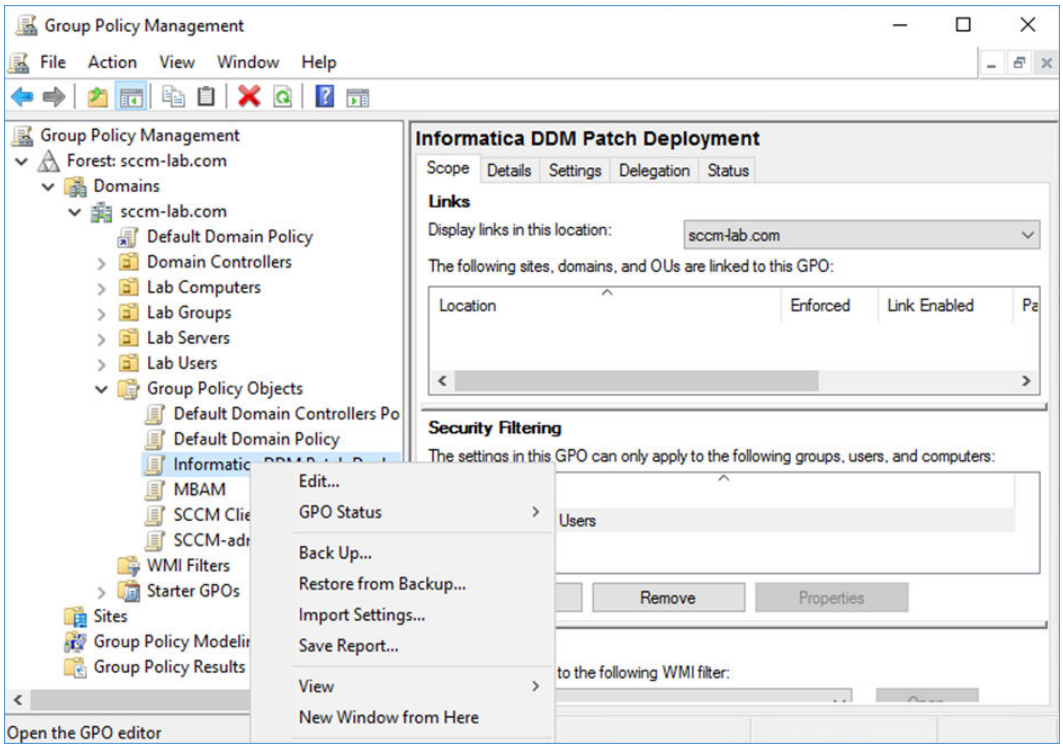


The following image shows the **New GPO** dialog box:



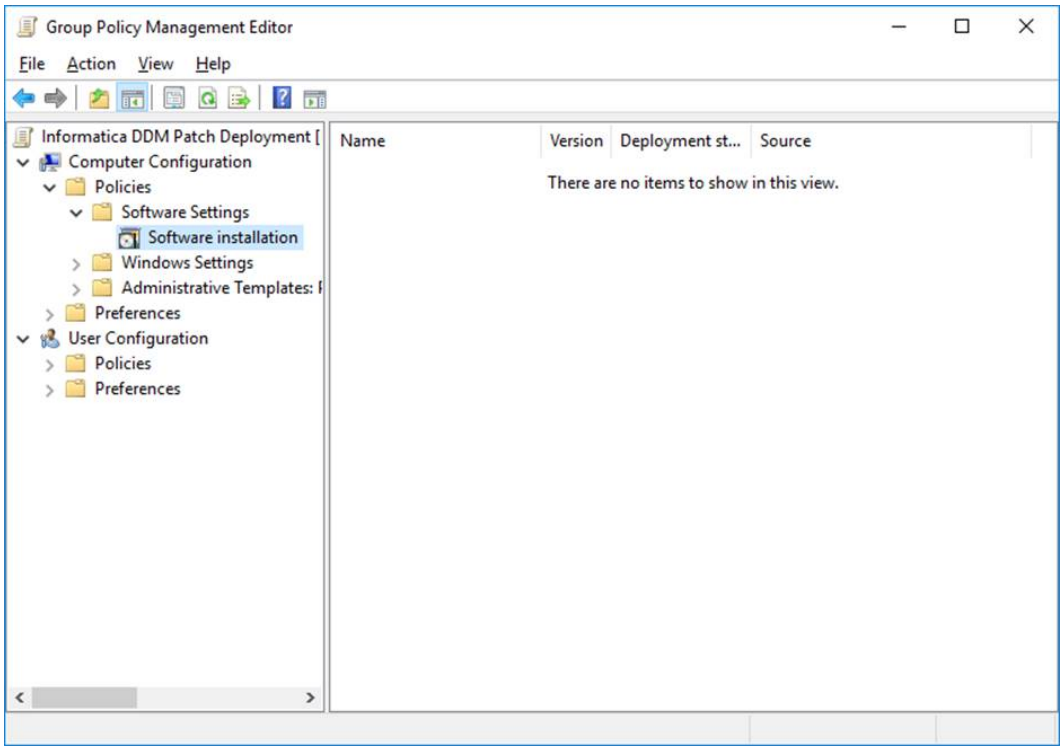
4. Right-click the created object and click **Edit**.

The following image shows the object **Edit** option:



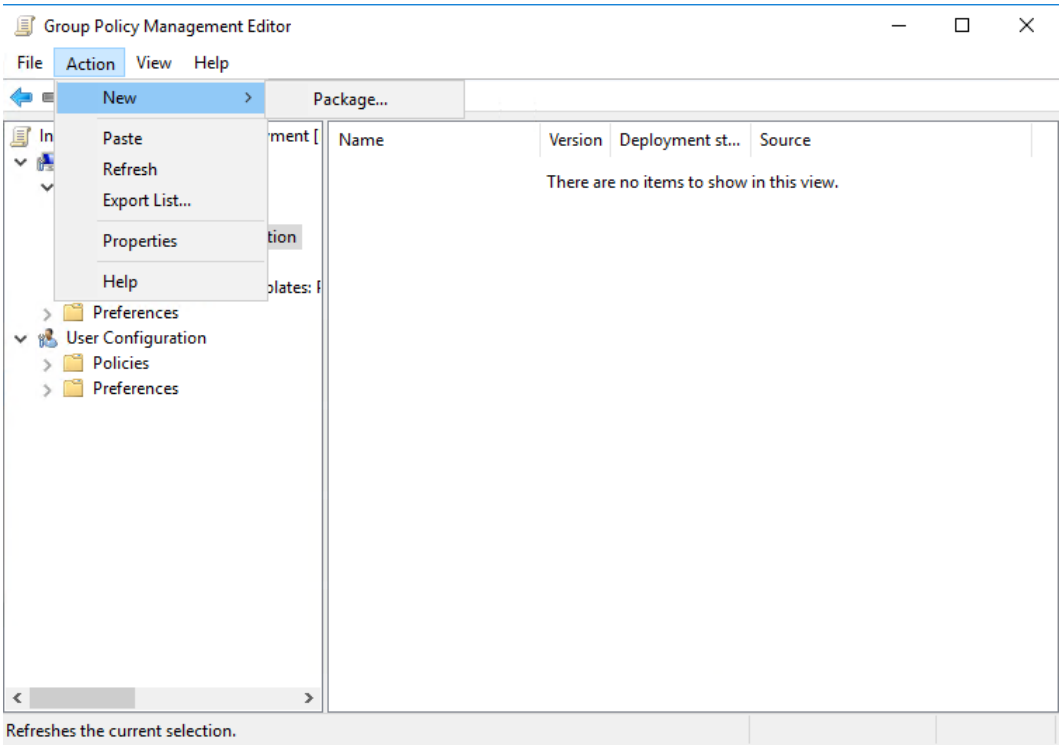
5. Navigate to **Computer Configuration** > **Policies** > **Software Settings** > **Software installation**.

   The following image shows the Software installation option selected on the Group Policy Management Editor:



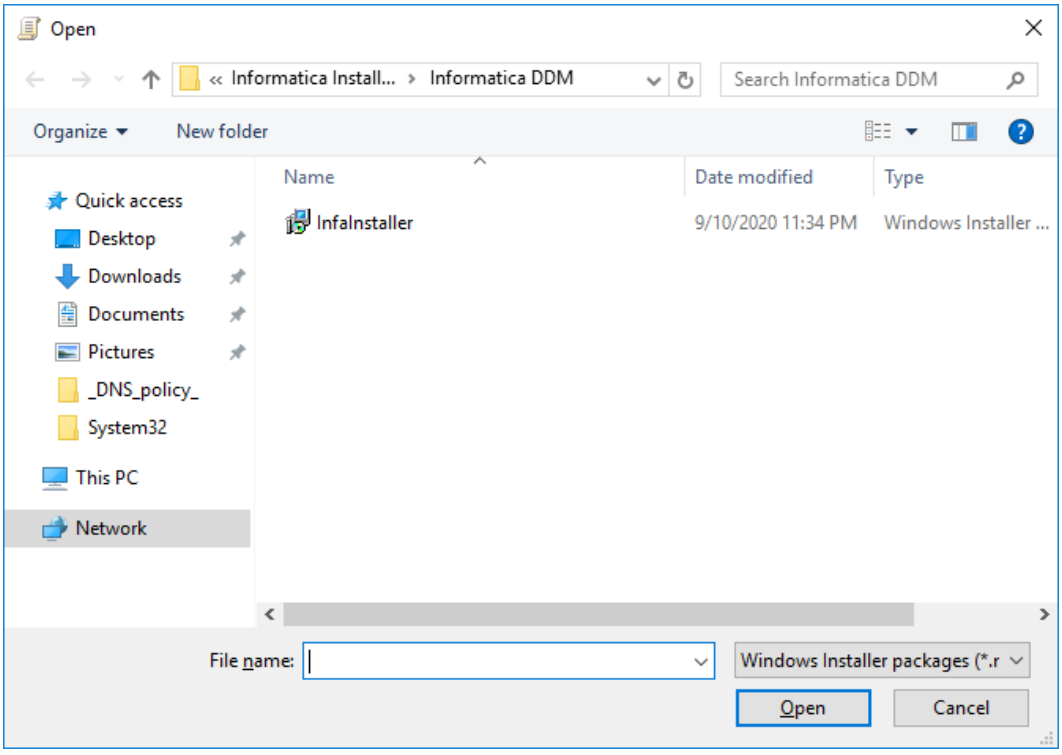
6. Click **Action** > **New** > **Package**

The following image shows the new package creation option:



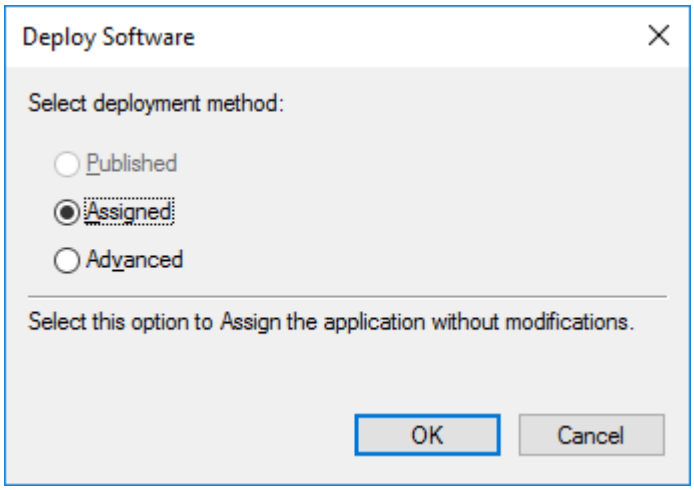
7. Navigate to the network folder that contains the installer package, click the page and then click **Open**.

The following image shows the search window:



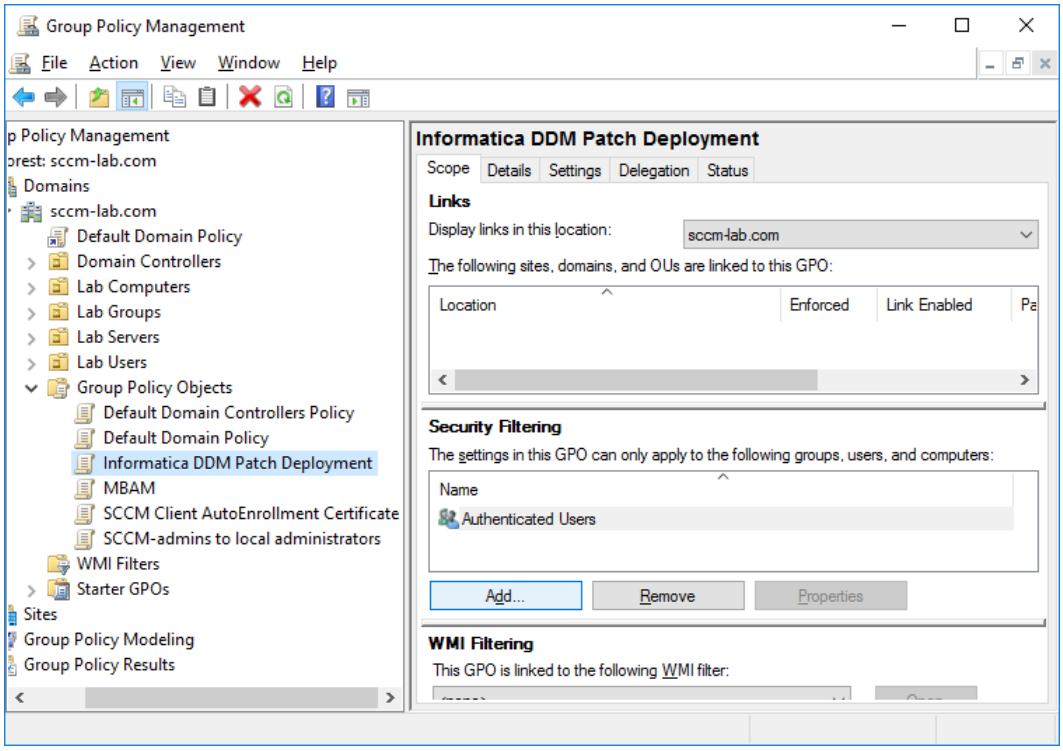
8. Choose the required deployment method and click **OK** in the **Deploy Software** dialog box.

The following image shows the **Deploy Software** window:



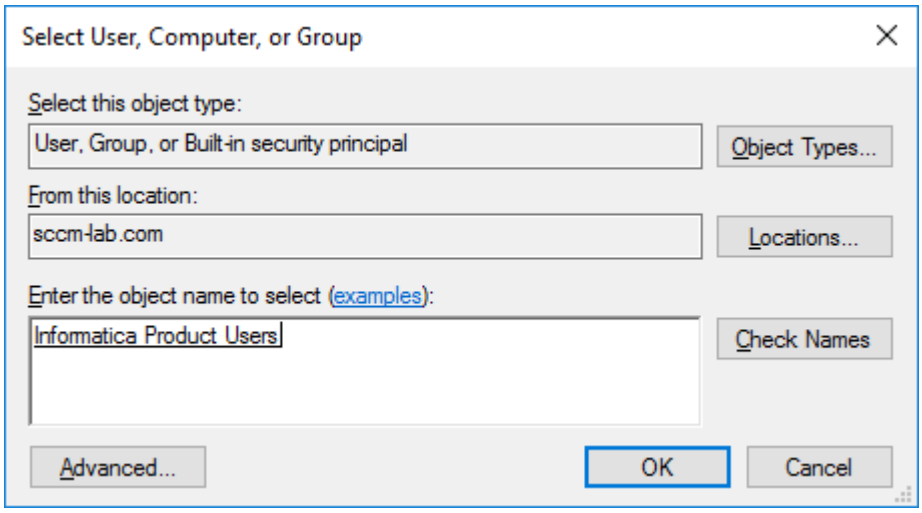
9. Close the Group Policy Management Editor.
10. To deploy the installer package to a specific group of users, select the object in the panel on the left and click **Add** on the **Security Filtering** tab on the right.

    The following image shows the **Security Filtering** tab of the group policy object:



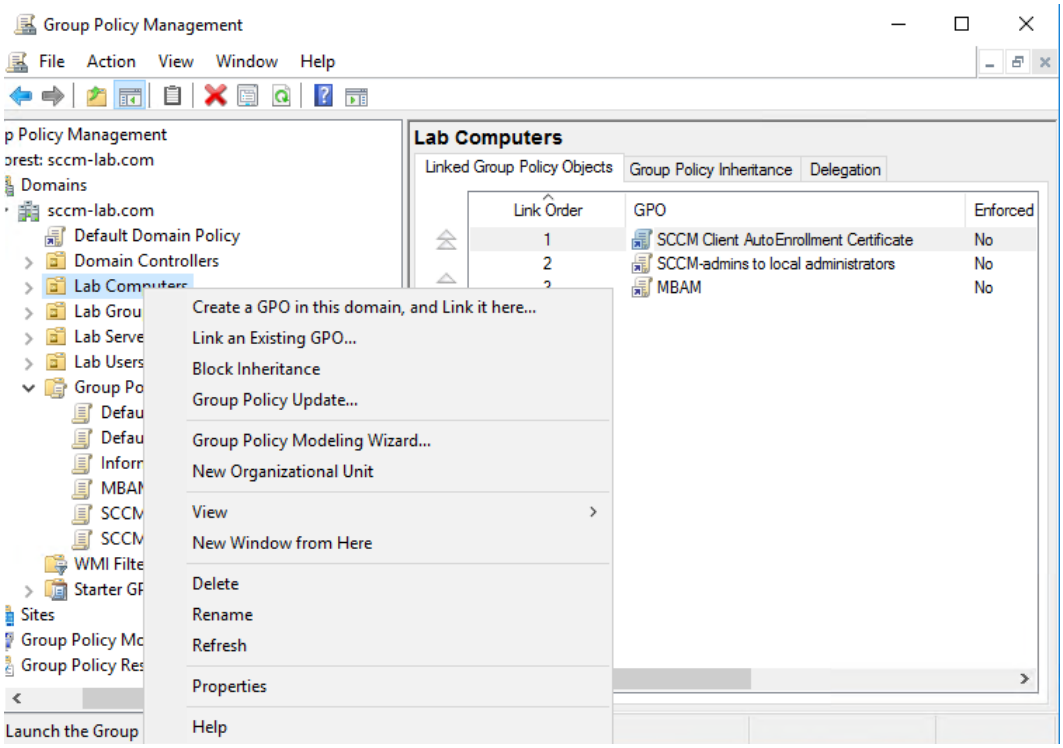
11. Enter the domain group that includes the target systems or users and click **OK**.

The following image shows the filter addition options window:



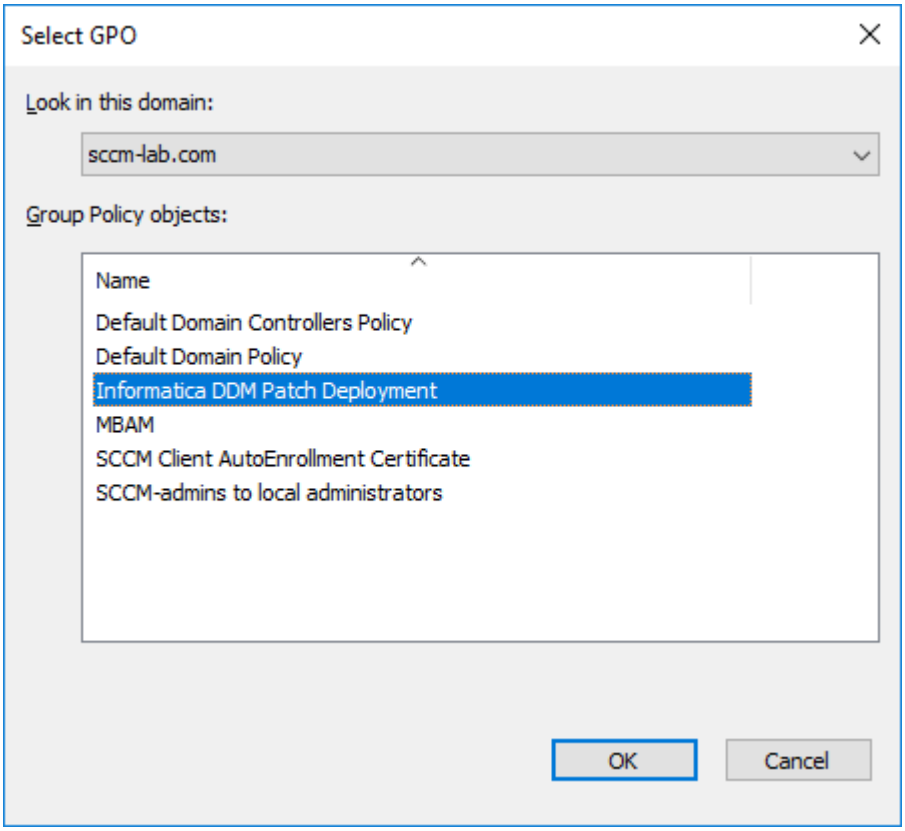
12. Optionally, you can modify filters such as Exclusion and WMI Filtering.
13. To link the object to an organization unit, right-click the organization unit and click **Link an existing GPO**.

    The following image shows the organization unit options:



14. On the **Select GPO** dialog box, select the Group Policy Object created and click **OK**.

    The following image shows the **Select GPO** dialog box:



    The package deploys the next time the system or user checks for group policy update.

For more information, see the Microsoft product documentation.

Author

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